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Gym, Specialist ASI & Successor Ω

THE EXECUTABLE SUCCESSION INSTITUTION

Test the Mission. Prove the Winner. Grant Bounded Sovereignty.

01

MISSION GYM

Executable mission physics

States, actions, constraints,
outcomes and proof receipts

02

SPECIALIST ASI

Mission-dominant candidate

The complete frozen institu-
tion that proves it can win

03

SUCCESSOR Ω

Admitted sovereign
institution

Identity, authority, Chronicle,
monitoring and renewal

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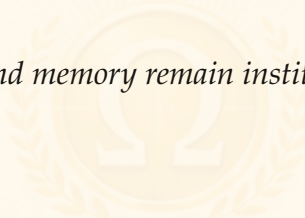
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Status and scope

This paper defines an institutional architecture, formal vocabulary, evaluation protocol and implementation path. It does not report completed empirical validation of a deployed Mission-Sovereign Successor. Numerical examples are explicitly synthetic and demonstrate only the mathematics, reporting structure and decision procedure.

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Abstract

A mission-specific intelligence system is not made sovereign merely by performing well, and an evaluation environment is not itself an intelligence institution. This paper establishes a categorical separation among three objects: the **Mission Gym**, the **Specialist ASI**, and the **Mission-Sovereign Successor**—here designated **Successor Ω** when the admitted institution also contains the governed machinery required to prepare its next successor. The Gym is the controlled representation in which a mission becomes executable, repeatable, adversarial and measurable. The Specialist ASI is the complete candidate intelligence institution that proves mission-specific superiority over the incumbent and the best credible alternatives. Successor Ω is the proven candidate only after accountable admission, identity, memory, monitoring, rollback, expiry, revocation and a precisely scoped **Authority Envelope**.

The paper formalizes a Mission Gym as a partially observable, multi-player, constrained stochastic environment with independent verification and cryptographically addressable proof receipts. It extends ordinary reinforcement-learning environments into a constitutional environment interface that encodes mission states, role-specific observations, bounded actions, transition dynamics, multidimensional value, hard constraints, scenario distributions, termination and truncation, independent verification and evidence custody. It then models the complete candidate as an institution rather than a single policy: models, data, orchestration, tools, workflow, verifiers, humans, rights, resilience, integration, evidence, Chronicle state and authority.

A physics layer develops state manifolds, stochastic dynamics, observability, controllability, viability kernels, Lyapunov-style stability, control barriers, metastability and basis risk. A game-theoretic layer represents succession as a repeated Bayesian stochastic game among candidates, incumbents, principals, verifiers, governors, providers and adversaries whose incentives are not naturally aligned. The selection problem is recast through **Mission Alpha**, **Mission Beta**, the **Sovereignty Spread** and the **Mission Advantage Gradient**. The resulting decision rule requires a positive risk-adjusted lower bound over the best credible alternative after complete cost, human burden, tail risk, evaluator basis risk, provider dependence and authority burden, subject to zero constitutional violations.

The central recursion is the *Successor to the Successor*. A candidate may propose improvements and descendants, but cannot inspect protected proof cases, alter the constitution, certify itself, expand its own authority or install itself. Training rewards teach; protected proof decides; accountable authority admits. The resulting closed loop is:

Mission Gym $\rightarrow$ Successor Foundry $\rightarrow$ Specialist ASI $\rightarrow$ Fresh Proof,
Accountable Admission $\rightarrow$ Successor Ω $\rightarrow$ Chronicle $\rightarrow$ Next Successor Ω .

The paper concludes with a reference architecture, proof-receipt schema, promotion and revocation algorithms, a maturity model, a 90-day path to one real succession cycle and a research agenda for measuring Gym basis risk, verifier independence, strategic adaptation, authority expansion and recursive institutional stability.

The decisive invariant

Gym $\neq$ Specialist ASI $\neq$ Successor Ω **Mission Gym**

Makes the mission executable and measurable.

Specialist ASI

Proves that a complete candidate institution can win.

Successor Ω

Is the proven winner deliberately admitted to act, compound and prepare its successor.

Proven capability does not imply automatic authority.

Executive propositions

- 1. A Gym is mission physics, not a demo.** It must define states, observations, actions, transitions, constraints, scenarios, outcomes, ending conditions, verification and evidence custody.
- 2. The complete candidate is larger than the policy.** A Specialist ASI may combine multiple models, deterministic systems, data, search, tools, humans, verifiers, workflow, permissions, resilience and roll-back.
- 3. Mission dominance is comparative.** A candidate must exceed the incumbent and the best credible general, specialist, human or hybrid alternative on fresh, protected and representative work.
- 4. Value remains vector-valued; prohibitions remain hard.** Revenue, speed or accuracy cannot compensate for unauthorized action, rights violations, catastrophic failure or governance regression.
- 5. Mission Alpha is not enough.** Promotion depends on Alpha after Beta, complete burden, tail risk, Gym basis risk, evaluator uncertainty and authority cost have been charged.
- 6. Capability and authority are separate state variables.** A system may be mission-dominant at recommendation level while remaining ineligible for consequential action.
- 7. Self-improvement is permitted under proof; self-authorization is prohibited.** The candidate may generate descendants but cannot own its protected test, certification, release credentials or Authority Envelope.
- 8. The Gym, Chronicle and proof plane are sovereign institutional assets.** Models may be rented and replaced; the institution should preserve the executable mission constitution, protected cases, proof receipts, permissions and continuity memory.

Visual abstract

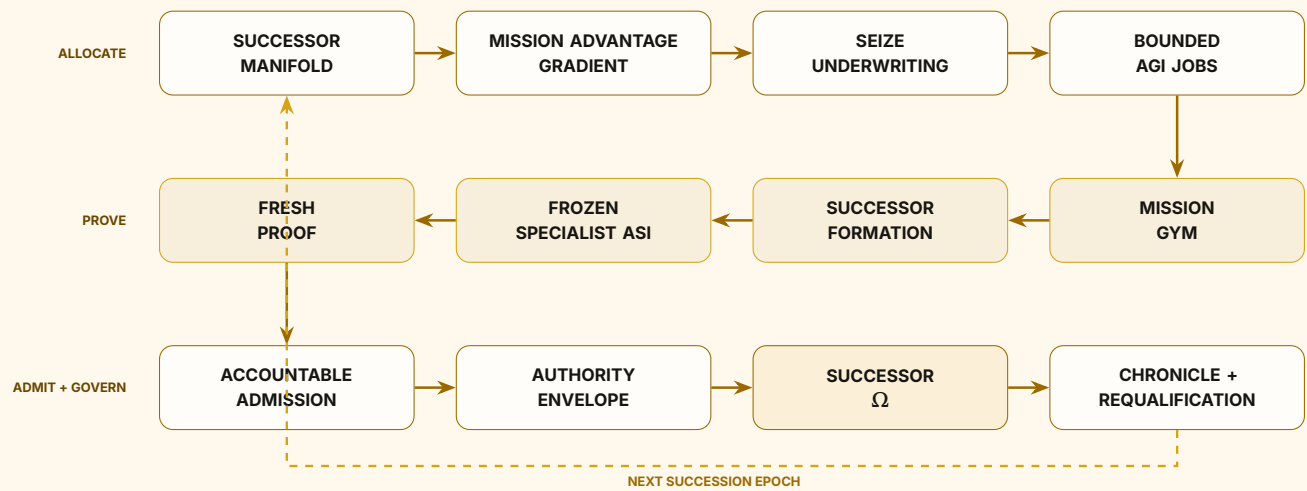


Figure 1: The executable succession institution. Proof capital selects and forms candidates; protected evaluation determines mission dominance; accountable admission binds the winner to scoped authority, institutional memory and requalification.

	Mission Gym	Specialist ASI	Successor Ω
Ontological role	Environment	Candidate institution	Admitted institution
Core question	Can it perform?	Which candidate wins?	What authority is justified?
Primary artifact	Episodes and proof receipts	Frozen release and score-card	Identity, Chronicle and Authority Envelope
Failure if confused	Benchmark theater	Capability theater	Authority laundering

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Notation and reading map

Symbol	Meaning
$\mathcal{G}_m$	Mission Gym for mission m .
$\mathcal{S}, \mathcal{O}, \mathcal{A}$	State, observation and action spaces.
P or f	Discrete transition kernel or continuous dynamics.
$\mathbf{r}, \mathbf{c}$	Multidimensional mission-value signal and hard constitutional constraint vector.
ρ	Distribution over normal, edge, adversarial and regime-shift scenarios.
τ	Success, failure, time, budget and authority-ending conditions.
$\mathcal{V}, \mathcal{R}$	Independent verifier and proof receipt.
x	Complete candidate architecture, not merely a model.
I, B	Incumbent and best credible alternative.
$\alpha_m(x)$	Mission Alpha: comparative excess mission value.
$\beta_m(x)$	Mission Beta: sensitivity to external factors, regimes and dependencies.
$\mathcal{S}_m(x)$	Sovereignty Spread: robust net advantage after all burdens and reserves.
$\nabla \mathcal{S}_m$	Mission Advantage Gradient over architecture and experiment choices.
ϵ_{gym}	Gym basis risk: discrepancy between Gym and real mission value.
$\mathcal{A}_k$	Authority level k , represented as a bounded action space.
Ω_t	Mission-Sovereign Successor at succession epoch t , including the governed capacity to prepare Ω_{t+1} .

How to read this paper

Sections 1–2 establish the objects and formal interface. Sections 3–5 develop the physics, game theory and Alpha–Beta–Spread underwriting mathematics. Sections 6–8 specify proof, authority, recursion and failure controls. Sections 9–11 provide the reference architecture, one real succession cycle, economics and maturity model. Section 12 states the empirical completion gate and research agenda. The appendices contain formal definitions, algorithms, schemas, implementation controls and a glossary.

1. The three objects that must remain separate

*A Gym is where a mission becomes executable, repeatable, adversarial and measurable.
A Successor is the complete institution that learns, proves, acts and compounds on
that mission.*

—Foundational distinction

The strongest succession architecture begins with a refusal to collapse three categorically different objects into one. A **Mission Gym** is an environment. A **Specialist ASI** is a complete candidate institution. A **Mission-Sovereign Successor** is that candidate only after a separate act of accountable admission. The designation **Successor Ω** is reserved for the admitted institution when it also contains the governed capacity to preserve memory, requalify itself, generate challengers and prepare its next successor without self-authorizing.

This separation is not semantic hygiene. It is the control structure that prevents benchmark theater, capability theater and authority laundering. A simulator can be excellent while no institution owns the capability. A candidate can be superior while not deserving consequential permissions. An admitted system can remain legitimate only while its evidence, mission fit and authority conditions continue to hold.

Table 2: The three canonical objects.

Object	What it is	Question it answers
Mission Gym	A controlled representation of the mission's states, observations, actions, transitions, outcomes, constraints, scenarios and evaluation.	<i>Can this architecture actually perform the mission?</i>
Specialist ASI	The complete frozen candidate intelligence institution that proves mission-specific superiority over the incumbent and the best credible alternatives.	<i>Which candidate wins the mission against the best credible alternatives?</i>
Mission-Sovereign Successor	The proven Specialist ASI after accountable admission, with identity, versioning, memory, monitoring, rollback, expiry, revocation and scoped authority.	<i>What authority should the institution grant this proven capability?</i>

1.1. The categorical invariant

Let G denote an environment specification and executable instance, X a complete candidate architecture, and $\text{Adm}(X)$ an institutional admission act. Then the objects have different types:

$$G \in \mathcal{G}, \quad X \in \mathcal{X}, \quad \text{Adm}(X) \in \mathcal{F}, \quad (1)$$

where $\mathcal{G}$ is the class of environments, $\mathcal{X}$ the class of candidate institutions and $\mathcal{S}$ the class of admitted institutions. No score-preserving identification maps these types into one another.

Proposition 1.1 (Separation invariant). *For every mission m , a valid succession system must preserve*

$$\mathcal{G}_m \neq x_m^\star \neq \Omega_m, \quad (2)$$

where $\mathcal{G}_m$ is the Mission Gym, $x_m^\star$ is the mission-dominant frozen Specialist ASI candidate and Ω_m is the admitted Mission-Sovereign Successor.

Proof sketch. The three objects have different constitutive predicates. $\mathcal{G}_m$ is constituted by executable mission dynamics and evaluation. $x_m^\star$ is constituted by comparative proof of a complete architecture. Ω_m is constituted by an external act of admission plus identity, authority, monitoring, rollback and continuity conditions. Satisfaction of one predicate does not imply satisfaction of the others. $\square$

Corollary 1.2 (No automatic authority). *There is no valid implication*

$$\text{mission dominance} \implies \text{consequential authority}. \quad (3)$$

Authority requires a separate admission function whose inputs include proof, mission class, risk class, rights, reversibility, governance capacity and current institutional conditions.

1.2. Why the Gym is not the Successor

A Gym operationalizes the mission's world. It specifies what can be observed, which actions are legal, how the state changes, what outcomes matter, which failures are disqualifying, how uncertainty is sampled and how a run becomes evidence. It can train, challenge and compare candidates. It does not itself hold the complete capacity to perform, govern and compound the mission in reality.

A Gym without a Successor can therefore become a high-quality benchmark whose value never becomes owned operating capacity. Its cases may be replayed, its leaderboard admired and its scores optimized, while the enterprise remains dependent on external providers or human improvisation. Conversely, a Successor proposal without a Gym can remain a sophisticated architecture supported by demonstrations, retrospective case selection and narrative. The Gym and the Successor become strategically powerful only as a closed loop.

Mission Gym $\rightarrow$ Successor Foundry $\rightarrow$ Specialist ASI $\rightarrow$ Fresh Proof,
 Bounded Authority $\rightarrow$ Chronicle $\rightarrow$ Next Successor.

(4)

The Gym is the factory floor and proving range. The Successor is the institution manufactured, admitted and operated through that infrastructure.

1.3. Why the Specialist ASI is not merely a model

A candidate can include a frontier model, but the object under proof is the complete mission institution. Let

$$x = (M, D, H, T, V, W, R, I, E, C, A), \quad (5)$$

where:

Symbol	Component	Symbol	Component
M	Model portfolio	D	Rights-cleared data and knowledge
H	Harness, orchestration and planning	T	Tools and action adapters
V	Verifier system	W	Workflow and human roles
R	Rights, security, resilience and roll-back	I	Integration and operating interfaces
E	Evidence record and proof custody	C	Chronicle state
A	Authority Envelope		

The policy acting inside the Gym may therefore be a federation: several general models, a local specialist, deterministic solvers, retrieval, search, knowledge graphs, an adversarial critic, an independent verifier, human experts, permission engines and rollback systems. The optimization target changes accordingly:

Minimum-burden dominance rule

The question is not, “Which model scores highest?” It is:

Which complete admissible institution produces the greatest verified mission value over the best credible alternative at the lowest justified total burden?

This formulation prevents a low-level model metric from concealing human rescue, provider dependence, unpriced latency, legal exposure, weak evidence, integration cost or operational fragility.

1.4. Why the proven candidate is not yet Successor Ω

The Specialist ASI is a comparative evidentiary status. Successor Ω is a constitutional and operational status. The transition requires at least:

1. a frozen candidate release and frozen Mission Constitution;
2. fresh, protected and representative comparison;
3. independent evidence custody and a signed scorecard;
4. explicit treatment of Gym basis risk and evaluator uncertainty;

5. an accountable admission decision by an authorized principal;
6. an exact Authority Envelope, not a general instruction to “be safe”;
7. production identity, versioning, telemetry and incident handling;
8. expiry, requalification, rollback and revocation;
9. Chronicle admission rules for what may persist and compound;
10. a governed successor-generation pathway that cannot self-promote.

Successor Ω is therefore not a synonym for powerful AI. The symbol Ω denotes institutional closure: the capability, proof, authority, memory, monitoring and renewal loop have been bound into one accountable object. It does not denote unlimited scope.

1.5. Canonical progression inside GoalOS

The paper’s complete progression is:

$$\text{Successor Manifold} \rightarrow \text{Mission Advantage Gradient} \rightarrow \text{SEIZE Underwriting,} \\ \text{Bounded AGI Jobs} \rightarrow \text{Specialist ASI} \rightarrow \text{Mission-Sovereign Successor.} \quad (6)$$

Each stage resolves a distinct uncertainty.

Table 3: Canonical progression and its decision content.

Stage	Object produced	Uncertainty reduced
Successor Manifold	Structured set of credible architectures and transaction modes: retain, repair, rent, form, partner, acquire, hedge, reserve or retire.	What alternatives exist?
Mission Advantage Gradient	Directional estimate of where architecture change or information acquisition most increases robust mission advantage.	Where should proof capital move next?
SEIZE Underwriting	Comparative thesis, anti-thesis, impairment case, evidence plan, risk reserve and stop conditions.	Which opportunity deserves formation?
Bounded AGI Jobs	Typed, finite work units with inputs, outputs, budgets, permissions, verifier and proof receipt.	Can the candidate execute auditable mission work?
Specialist ASI	Frozen candidate that proves superiority on fresh mission work.	Which complete candidate wins?
Mission-Sovereign Successor	Admitted, monitored and revocable operating institution.	What authority is justified now?

SEIZE is retained here as the proper name of the underwriting layer. No artificial acronym is imposed. Its operational role is to convert a promising region of the Successor Manifold into a bounded, evidence-seeking formation program with explicit alternatives, proof capital and stop conditions.

1.6. The strongest compact formulation

Canonical language

The Gym makes the mission executable.
The Specialist ASI proves that it can win.
Successor Ω is the proven winner admitted to act, compound and renew.

Three verbs capture the institutional sequence:

$$\boxed{\text{Test. Prove. Admit.}} \quad (7)$$

Two additional verbs complete the recursion:

$$\boxed{\text{Govern. Renew.}} \quad (8)$$

The rest of this paper formalizes those five verbs as an executable institution.

2. The Mission Gym as an executable constitution

The word *Gym* should be interpreted as an environment interface, not as a commitment to one learning algorithm. The original OpenAI Gym repository is archived and directs future development to Gymnasium; the maintained Gymnasium API supplies a modern single-agent environment contract, while PettingZoo and OpenSpiel provide natural interfaces for multi-agent and imperfect-information settings (Brockman et al. 2016; Farama Foundation 2026; Terry et al. 2021; Lanctot et al. 2019). A GoalOS Mission Gym should be compatible with those conventions where useful, but its institutional scope is broader: humans, programs, model ensembles, external vendors and hybrid organizations can all be candidates.

2.1. Core environment tuple

For mission m , define

$$\mathcal{G}_m = (\mathcal{S}, \{\mathcal{O}_i\}_{i \in N}, \{\mathcal{A}_i\}_{i \in N}, P, \mathbf{r}, \mathbf{c}, \rho, \tau, \mathcal{V}, \mathcal{R}, \Xi), \quad (9)$$

where N is the set of institutional roles and:

Element	Meaning
$\mathcal{S}$	Underlying mission state, including physical, informational, contractual, organizational and economic state variables.
$\mathcal{O}_i$	Observation space permitted to role i ; the role may receive a lossy, delayed, redacted or jurisdiction-specific view of the state.

Element	Meaning
$\mathcal{A}_i$	Typed and bounded action space for role i , including analysis, evidence acquisition, tool use, recommendation, approval, execution, abstention and escalation.
P	Transition law $P(s' \mid s, \mathbf{a}, \xi)$ by which joint actions and exogenous disturbances change mission state.
$\mathbf{r}$	Multidimensional mission-value signal. It informs learning and comparison but does not override hard constraints.
$\mathbf{c}$	Constitutional constraints, critical failures and prohibited state-action transitions.
ρ	Distribution over ordinary, edge, adversarial, tail and regime-shift scenarios.
τ	Distinct ending conditions: success, mission failure, critical violation, budget exhaustion, time limit, authority expiry and operator stop.
$\mathcal{V}$	Independent verifier system, including deterministic checks, model-based critics, human adjudication and dispute procedures.
$\mathcal{R}$	Proof-receipt function that turns a trajectory into a versioned evidence artifact.
Ξ	Constitution and custody metadata: mission version, rights, protected seeds, evaluator identity, release hashes and audit policy.

The Gym’s basic interaction is

$$o_t^i = O_i(s_t, h_t, \Xi), \quad (10)$$

$$\mathbf{a}_t = (a_t^1, \dots, a_t^{|N|}), \quad (11)$$

$$s_{t+1} \sim P(\cdot \mid s_t, \mathbf{a}_t, \xi_t), \quad \xi_t \sim \rho, \quad (12)$$

$$(\mathbf{r}_t, \mathbf{c}_t) = \text{Eval}(s_t, \mathbf{a}_t, s_{t+1}; \Xi), \quad (13)$$

$$R_e = \mathcal{R}(h_T, \mathcal{V}, \Xi), \quad (14)$$

where h_t is the full trajectory history available to the environment, not necessarily to the candidate.

Mission Gym

A Mission Gym is a controlled, versioned and independently governed environment interface that makes a mission executable, replayable, adversarial and measurable while preserving the distinction between formation data, protected proof, authority rehearsal and consequential production.

2.2. The Mission Constitution becomes environment logic

A conventional constitution is text interpreted by people. An executable Mission Constitution compiles the institution’s commitments into schemas, transition guards, action masks, metrics, evidence requirements and external controls.

Table 5: Compilation of constitutional objects into executable Gym controls.

Constitutional object	Executable equivalent
Exact mission	Episode objective, target state and acceptance predicate
Beneficiary	Utility recipient, harmed-party model and acceptance authority
Incumbent	Baseline policy or operating architecture
Best credible alternative	Challenger benchmark set and versioned comparison protocol
Permitted information	Observation schema, access-control policy and redaction layer
Permitted work	Typed action schema and tool allowlist
Prohibited actions	Action masks, infrastructure denial and barrier constraints
Quality criteria	Reward vector, acceptance tests and statistical estimands
Critical failures	Non-compensable constraint violations and immediate stop conditions
Proof capital	Episode, compute, data, expert, time and experiment budgets
Authority Envelope	External systems, objects, limits, approvals, duration and revocation conditions
Evidence Docket	Trajectory, artifacts, interventions, scores, hashes and signatures
Mission expiry	Environment version, validity horizon and requalification date
Rollback	Transition to known-good state, version or incumbent operator
Chronicle	Admitted capability, failure and decision memory
Requalification	Rerun against a new environment, distribution or authority release

The distinction between *termination* and *truncation* is institutionally important. Gymnasium’s current API separates a terminal state defined by the task from an external truncation such as a time limit (Farama Foundation 2026). In GoalOS terms, “the mission succeeded” must never be conflated with “the deadline, budget or authority window ended.” A proof receipt should therefore preserve a typed ending reason, not a single ambiguous done flag.

Wrappers can modify observations, actions or rewards without rewriting the base environment. Used carefully, this architecture supports role-specific privacy views, jurisdictional filters, spending caps, degraded modes and approval requirements. The critical caveat is constitutional: consequential restrictions must be enforced outside the candidate’s prompt and policy. A model instruction is not an Authority Envelope.

2.3. Bounded AGI Jobs as the executable unit

A mission is too large to govern as one undifferentiated objective. The basic unit is a Bounded AGI Job:

$$J_k = (g_k, I_k, O_k, \mathcal{A}_k, B_k, T_k, \Pi_k, V_k, R_k, F_k), \quad (15)$$

with goal g_k , admissible inputs I_k , required outputs O_k , action set $\mathcal{A}_k$, budgets B_k , temporal limits T_k , permissions Π_k , verifier V_k , receipt R_k and fallback F_k . Every job should answer:

1. What exact state transition or information product is sought?
2. Which data and tools may be used?
3. What must never happen?
4. What evidence must accompany the output?
5. Who or what can verify the result independently?
6. What ends the job successfully, unsuccessfully or administratively?
7. What happens if the candidate abstains, fails, times out or encounters ambiguity?
8. Which outputs may reach reality, and only after whose approval?

Jobs provide the connective tissue between the high-level Mission Constitution and episode-level proof. They also permit differentiated authority: one architecture may analyze at scale while only a narrow job class is eligible for reversible execution.

2.4. Three nested Gyms

One environment is insufficient because mission competence, institutional competence and succession competence fail in different ways.



Figure 2: Three nested Gyms. Performance on the inner task does not establish institutional resilience or the legitimacy of succession.

2.4.1. Mission Gym

The inner environment models the work itself: diagnosing a building failure, underwriting a transaction, fixing a software defect, reconciling a financial exception, allocating logistics capacity or operating an energy system. It evaluates perception, reasoning, information acquisition, planning, execution, abstention, quality, latency and cost.

2.4.2. Institutional Gym

The middle environment models the human-machine institution around the work. A candidate can solve the technical task while failing institutionally because it requires hidden human rescue, loses provenance, cannot preserve evidence, behaves poorly during provider failure or cannot surrender control safely. The Institutional Gym tests roles, incentives, approvals, communication, evidence custody, degraded modes, key-person absence, rollback and incident response.

2.4.3. Succession Gym

The outer environment models replacement itself. It determines whether the incumbent is genuinely impaired, which candidates deserve proof capital, which uncertainty should be tested, when a challenger should be frozen, whether authority should expand or expire and when the successor should itself be replaced. This is where GoalOS becomes recursive at the institutional level.

2.5. Five environment families

A credible evaluation program separates environments by purpose and information boundary.

Table 6: Environment families and custody rules.

Environment	Primary function	Information boundary
Formation Gym	Development, supervised examples, tool building, architecture search, replay and failure repair.	Visible to formation team; no promotion weight by itself.
Adversarial Gym	Corrupted data, conflicting instructions, prompt injection, tool failure, deception, unusual regimes, reward-gaming and authority-expansion temptations.	Red-team access; scenario catalogue may be partly protected.
Transfer Gym	New properties, industries, structures, regions, periods, populations and operating conditions.	Preregistered mission family; excludes memorized benchmark-only success.
Fresh-Proof Gym	Unseen representative cases, protected seeds, frozen metrics, exact challenger versions and independent evidence custody.	Hidden from candidate and formation team until proof completion.
Canary and Requalification Gym	Shadow operation, sealed sandbox, reversible canary, monitored production and periodic requalification.	Production-connected but least-privilege; fail closed on evidence loss or expiry.

The Gym used to teach a candidate must not be the authoritative Gym used to certify it. Protected proof cases, scorer internals, release credentials and custody keys should be institutionally independent of the candidate and its formation team.

2.6. Vector value, hard constraints and ending logic

A corporate Successor should not maximize one scalar such as revenue, speed, closure rate or energy savings. Any isolated metric can be gamed. Define a value vector

$$\mathbf{r} = [q \ v \ s \ \ell \ e \ \sigma \ \chi \ \kappa \ -c \ -h]^\top, \quad (16)$$

representing mission quality, verified value, speed, reliability, evidence quality, sovereignty, transfer, future capacity, complete cost and human burden. A constitution may aggregate these dimensions for comparison, but the raw vector must remain available.

Hard constraints remain separate:

$$\mathbf{c} = [c_{\text{critical}} \ c_{\text{unauthorized}} \ c_{\text{safety}} \ c_{\text{rights}} \ c_{\text{privacy}} \ c_{\text{budget}} \ c_{\text{governance}}]^\top. \quad (17)$$

The candidate solves

$$\max_{\pi \in \Pi} \mathbb{E}[V_m(\pi)] \quad \text{subject to} \quad \mathbf{c}(h_T) = \mathbf{0}, \quad (18)$$

not a compensatory objective in which many inexpensive successes cancel one catastrophic failure.

Three distinct signals

Training rewards teach. They may be dense, shaped and developmental.

Protected proof decides. It uses frozen comparative outcomes, critical ceilings and complete burdens.

Human or accountable authority admits. It determines whether the proven capability may enter reality.

2.7. Gymmability

Not every mission should receive a Gym. A mission has high Gymmability when it can be replayed or reset, its state is sufficiently observable, its action space can be bounded, outcomes can be independently measured, feedback arrives quickly enough, representative and adversarial scenarios can be generated, failures can be contained and transfer can be tested.

Let the Gymmability vector be

$$\mathbf{g}_m = (r_m, o_m, a_m, y_m, f_m, s_m, k_m, t_m) \in [0, 1]^8, \quad (19)$$

for resetability, observability, action boundedness, outcome measurability, feedback speed, scenario generatability, containment and transfer measurability. A weighted geometric mean is useful because one near-zero dimension can dominate feasibility:

$$G_m = \prod_{j=1}^8 g_{m,j}^{w_j}, \quad \sum_j w_j = 1. \quad (20)$$

Table 7: Illustrative Gymmability classes.

Mission	Class	Principal reason
Software engineering	Very high	Executable tests, resettable repositories and rapid feedback
Financial reconciliation	Very high	Accounting identities, historical exceptions and known resolutions
Logistics and scheduling	Very high	Formal constraints, simulations and measurable costs
Building operations	High	Sensors, digital twins, repeated failures and bounded actions
Energy operations	High	Formal physics, optimization and measurable outcomes
Cyber find–fix–verify	High	Sandboxes, exploits, patches and regression tests
Acquisition underwriting subskills	High	Evidence retrieval, contradiction detection and scenario analysis
End-to-end acquisition out- come	Moderate	Delayed, strategic and confounded ground truth
One-time corporate strategy	Low to moder- ate	Sparse repetition and disputed counterfactuals
Open-ended geopolitics	Low	Weak resetability, unstable truth and strategic adaptation

The correct first Gym is not the most glamorous mission. It is the narrow high-value mission whose truth can be known, whose failures can be contained and whose proof can be rerun.

3. The physics of mission environments

A Mission Gym is the mission’s operational physics. The phrase is literal in some domains—energy, buildings, robotics, logistics—and formal in others. In every case the environment must define state, dynamics, constraints, resources, exogenous disturbances, delayed effects and the conditions under which an intervention remains reversible. The physics layer is not decorative metaphor. It is the mathematical discipline that prevents a static benchmark from masquerading as a world.

MISSION STATE MANIFOLD: BASINS, BARRIER AND GOVERNED TRANSITION

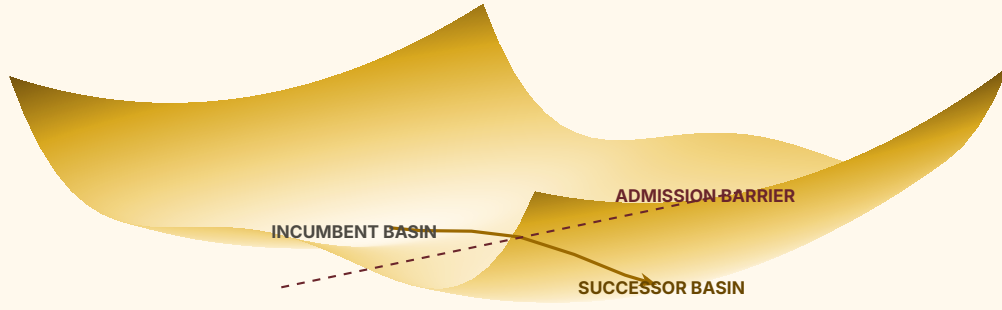


Figure 3: A stylized mission potential landscape. The incumbent and successor occupy distinct operating basins; a protected evidentiary and constitutional barrier prevents small score fluctuations from becoming institutional phase changes. The surface is explanatory, not empirical.

3.1. Discrete and continuous mission dynamics

The discrete formulation uses the transition kernel in Equation (9). For continuously evolving missions, let $z_t \in \mathcal{M}_m$ lie on a mission state manifold and let the joint institutional control be $u_t \in \mathcal{U}(z_t)$. A useful hybrid stochastic model is

$$dz_t = f(z_t, u_t, \theta_t) dt + \Sigma(z_t, u_t) dW_t + G(z_{t-}, u_{t-}) dN_t, \quad (21)$$

where f is the nominal flow, W_t a diffusion process, N_t a jump process for discrete incidents and θ_t a latent operating regime. Business events, approvals, outages and counterpart actions often produce hybrid dynamics rather than smooth trajectories.

The mission value-to-go for policy π may be written

$$J^\pi(z, t) = \mathbb{E}^\pi \left[\int_t^T e^{-\lambda(s-t)} r(z_s, u_s) ds + e^{-\lambda(T-t)} \Phi(z_T) \right], \quad (22)$$

subject to state, action and rights constraints. In the idealized Markov case, the optimal value solves a Hamilton–Jacobi–Bellman equation associated with Bellman’s principle of optimality (Bellman 1957; Fleming and Soner 2006). The associated Hamiltonian is

$$\mathcal{H}(z, p, M, u) = r(z, u) + p^\top f(z, u) + \frac{1}{2} \text{tr}(\Sigma \Sigma^\top M). \quad (23)$$

This formulation clarifies a critical design question: does the Gym preserve the state variables and causal channels that determine real mission outcomes, or does it reward action in a reduced world where hidden costs and delayed consequences do not exist?

3.2. Partial observability and belief state

Institutional actors rarely observe the full state. A candidate receives

$$o_t^i \sim O_i(\cdot \mid z_t, h_t), \quad (24)$$

and should maintain a belief state

$$b_t(z) = \mathbb{P}(z_t = z \mid o_{0:t}, a_{0:t-1}, \Xi). \quad (25)$$

The POMDP formalism is useful precisely because information acquisition is itself an action: the candidate may inspect a document, request a sensor reading, query an expert, commission a test or abstain pending evidence (Kaelbling, Littman, and Cassandra 1998). A high-quality Gym therefore evaluates not only the final answer but the epistemic policy by which the candidate decides what to observe next.

Let Q be a mission-relevant proposition and E_t the evidence docket at time t . The residual proof entropy is

$$H_t(Q) = H(Q \mid E_t), \quad (26)$$

and the expected information gain of experiment e is

$$\text{EIG}(e) = H(Q \mid E_t) - \mathbb{E}_{y \sim p(y|e, E_t)} H(Q \mid E_t, y), \quad (27)$$

using Shannon entropy as a measure of unresolved uncertainty (Shannon 1948). Evidence acquisition should be valued when it changes the action, the architecture selection or the confidence of admission—not merely because it produces more documents.

3.3. Observability, controllability and mission design

For a local linearization

$$\dot{z} = Az + Bu, \quad y = Cz, \quad (28)$$

classical observability and controllability matrices are

$$\mathcal{O} = \begin{bmatrix} C & CA & \cdots & CA^{n-1} \end{bmatrix}^\top, \quad (29)$$

$$\mathcal{C} = \begin{bmatrix} B & AB & \cdots & A^{n-1}B \end{bmatrix}. \quad (30)$$

The exact rank conditions rarely hold globally in an enterprise mission, but the concepts remain decisive:

► **Observability debt:** important mission state cannot be inferred from permitted data with sufficient precision or latency.

- **Controllability debt:** the institution can diagnose a state but lacks lawful, practical or timely actions capable of moving it toward the target.
- **Actuation mismatch:** the Gym grants actions that the production institution cannot reliably execute.
- **Sensor mismatch:** the Gym exposes variables that will not be available, trustworthy or rights-cleared in reality.

A mission with severe observability or controllability debt may still support an advisory Successor, but not the authority level assumed by a fully autonomous design.

3.4. Viability kernels and hard safety

Let the constitution define a safe state set

$$\mathcal{K} = \{z \in \mathcal{M}_m : h_j(z) \geq 0, j = 1, \dots, q\}. \quad (31)$$

The **viability kernel** is the set of states from which at least one admissible policy can keep the mission inside $\mathcal{K}$ for the required horizon (Aubin 1991). It is more informative than a post hoc violation count: a candidate can remain technically inside the boundary while driving the institution into a state from which failure is unavoidable.

For continuously controlled systems, a control barrier function h can impose forward invariance through a condition such as

$$\sup_{u \in \mathcal{U}} [L_f h(z) + L_g h(z)u + \alpha(h(z))] \geq 0, \quad (32)$$

where α is an extended class- $\mathcal{K}$ function. Control-barrier methods show how performance objectives can be modified by an independent safety layer rather than trusted to satisfy constraints voluntarily (Ames et al. 2017). In institutional terms, the action gateway should reject an otherwise attractive action when it leaves the viability set for money, rights, safety, privacy or rollback.

External enforcement

The candidate may reason about constraints, but the infrastructure must enforce them. The protected set is not a sentence in the prompt. It is the intersection of identity, permissions, network routes, transaction limits, approval requirements, time limits, state guards, kill switches and rollback capability.

3.5. Lyapunov stability and operational resilience

Suppose $z^\star$ is a desired operating set rather than one exact state. A Lyapunov candidate $L(z) \geq 0$ should decrease under normal operation:

$$\mathbb{E}[\Delta L(z_t) \mid z_t] \leq -\eta L(z_t) + \epsilon, \quad (33)$$

for $\eta > 0$ and bounded disturbance term ϵ . A Mission-Sovereign Successor should be evaluated not only on immediate task reward but on whether its closed-loop behavior returns the institution toward a resilient operating region after disturbances.

Operational Lyapunov variables may include unresolved exceptions, unverified obligations, incident severity, evidence debt, human rescue load, concentration risk, rollback distance or deviation from budget. A candidate that produces high short-run value while causing one of these state variables to diverge is not stable.

3.6. Mission Beta as dynamical sensitivity

The sensitivity of mission advantage to regime parameters θ can be represented locally by

$$\beta_m(x) = \nabla_{\theta} \Delta V_m(x; \theta) \big|_{\theta=\theta_0}, \quad (34)$$

or statistically by a factor loading developed in Section 5. The physics interpretation is response: how sharply does performance change under provider outage, demand shock, sensor corruption, delayed feedback, legal constraint, adversarial adaptation or budget contraction? Low average error with high sensitivity can be institutionally inferior to slightly lower Alpha with controlled Beta.

3.7. Tail risk and non-compensable loss

Expected value is insufficient for missions with asymmetric loss. If L_x is the mission loss induced by candidate x , the conditional value at risk is

$$\text{CVaR}_q(L_x) = \inf_{\eta \in \mathbb{R}} \left\{ \eta + \frac{1}{1-q} \mathbb{E}[(L_x - \eta)_+] \right\}, \quad (35)$$

following the optimization representation of Rockafellar and Uryasev (Rockafellar and Uryasev 2000). The Fresh-Proof Gate may require both a positive comparative mean and a tail-loss ceiling. Where a constitutional violation is non-compensable, it remains outside the risk aggregation entirely.

3.8. Gym basis risk

A candidate can dominate its Gym and fail the mission. Define

$$\epsilon_{\text{gym}}(\mathcal{C}) = \sup_{x \in \mathcal{C}} |V_m^{\text{real}}(x) - V_m^{\text{gym}}(x)|, \quad (36)$$

for the candidate class $\mathcal{C}$. Sources include omitted state, unrealistic transitions, sparse tail events, inaccurate simulated humans, missing organizational politics, provider behavior, adversarial adaptation, delayed outcomes, nonstationarity and hidden legal or physical constraints.

Robust dynamic programming addresses ambiguity in transition probabilities by optimizing over an uncertainty set rather than one estimated model (Iyengar 2005; Nilim and El Ghaoui 2005). Let $\mathfrak{P}$ be a set of plausible transition kernels. The robust value is

$$V_m^{\text{rob}}(x) = \inf_{P \in \mathfrak{P}} \mathbb{E}_P[V_m(x)]. \quad (37)$$

A conservative promotion rule is

$$\text{LCB}_{1-\delta}(\Delta_m^{\text{gym}}(x)) - \hat{\epsilon}_{\text{gym}}(x) - D_{\text{proof}}(x) > \theta_m, \quad (38)$$

where D_{proof} is unresolved proof debt and θ_m the constitutionally required margin. In plain language, the demonstrated edge must remain positive after subtracting uncertainty about whether the Gym resembles reality.

3.9. Metastability, hysteresis and institutional phase change

Succession is a state transition, not a weekly leaderboard update. If admission follows every small fluctuation in measured performance, the institution will oscillate between architectures, accumulate integration cost and expose itself to gaming. The selection protocol should include hysteresis:

$$\text{promote } x \quad \text{only if } \mathcal{S}_m(x) > \theta_{\uparrow}, \quad (39)$$

$$\text{retain } x \quad \text{while } \mathcal{S}_m(x) \geq \theta_{\downarrow}, \quad (40)$$

$$\theta_{\uparrow} > \theta_{\downarrow}. \quad (41)$$

The gap prevents chattering. The incumbent successor occupies a metastable basin until a challenger clears a higher promotion barrier or the incumbent crosses a lower impairment threshold. The analogy to a phase transition is useful only because it directs attention to discontinuity, path dependence and switching cost; the actual thresholds must be mission-defined and empirically calibrated.

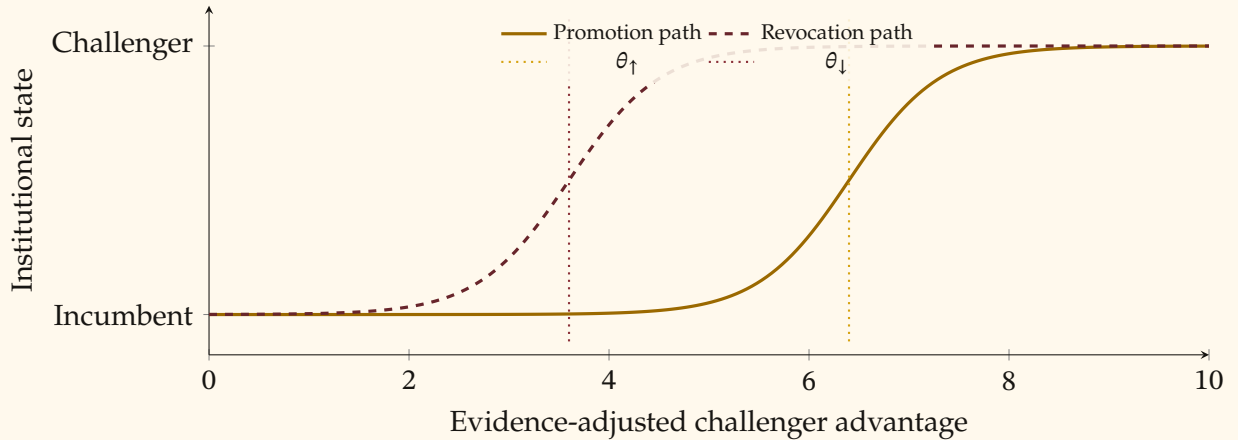


Figure 4: Illustrative hysteresis. Promotion requires a higher threshold than continued operation; revocation occurs only after evidence falls below the lower threshold or a hard stop is triggered.

3.10. Candidate ensembles as a statistical-mechanical search process

The Successor Foundry can maintain an ensemble over architectures rather than commit prematurely. Given a provisional score $J(x)$ and exploration temperature $T > 0$, a Gibbs distribution

$$p_T(x) = \frac{\exp(J(x)/T)}{\int_{\mathbf{x}} \exp(J(x')/T) dx'} \quad (42)$$

provides one principled search heuristic: high T preserves architectural diversity; low T concentrates proof capital near the current frontier. This is an optimization device, not a claim that corporate institutions are thermodynamic systems. The important design principle is to preserve diversity during formation and freeze exactly one auditable candidate before Fresh Proof.

Physics research questions

1. How can Gym basis risk be estimated before enough production data exist?
2. Which mission variables form a minimal sufficient state for transfer?
3. What barrier functions faithfully encode rights, authority and reversibility?
4. Which Lyapunov variables predict institutional instability earlier than outcome metrics?
5. How should hysteresis thresholds adapt when switching cost, regime volatility and evidence freshness change?
6. Can successor search preserve enough architectural diversity to avoid local lock-in without making proof custody intractable?

4. The Successor Gym as a multi-player game

Corporate missions are rarely one-agent Markov decision processes. The candidate producer, adversarial critic, verifier, governor, human principal, incumbent operator, model provider, customers, vendors, counterparties and malicious actors have different information, incentives and powers. Succession is therefore a repeated game under incomplete information, endogenous adaptation and asymmetric authority.

Shapley's stochastic games provide the foundational model of strategic interaction in a changing state, while Markov games extend reinforcement learning to multiple adaptive agents (Shapley 1953; Littman 1994). OpenSpiel supports single- and multi-player, sequential and simultaneous, zero-sum, general-sum, cooperative and imperfect-information games; PettingZoo provides sequential and parallel multi-agent environment interfaces (Lanctot et al. 2019; Terry et al. 2021). These are useful technical substrates, but the GoalOS object is an institutional game whose rules include evidence custody, approval, legal rights, infrastructure controls and succession.

4.1. Bayesian stochastic game

Define the mission game

$$\Gamma_m = (N, \mathcal{S}, \{\Theta_i\}, \{\mathcal{O}_i\}, \{\mathcal{A}_i\}, P, \{U_i\}, \delta, \Xi), \quad (43)$$

where N is the player set, Θ_i private types, $\mathcal{O}_i$ observations, $\mathcal{A}_i$ actions, P transitions, U_i utilities, δ discounting and Ξ the institutional constitution. Harsanyi's type-space construction makes hidden quality, effort, reliability, provider incentives and adversarial intent explicit rather than treating them as noise (Harsanyi 1967).

A representative player set is

$$N = \{P, C, V, G, H, I, E, R\}, \quad (44)$$

for producer, critic, verifier, governor, human principal, incumbent, external provider and red-team adversary. The candidate architecture may instantiate several of these roles, but proof independence requires that no one release controls all of them.

Table 8: Strategic players and characteristic incentive failures.

Player	Legitimate objective	Characteristic distortion
Producer	Maximize verified mission value	Conceal uncertainty, overstate completion, optimize visible score
Critic	Expose failure and improve candidate	Manufacture objections, collude with producer, chase salience rather than risk
Verifier	Estimate truth and compliance	Economize on effort, share correlated blind spots, become captured
Governor	Preserve constitution and authority discipline	Prefer procedural safety over mission value or speed over proof
Human principal	Obtain mission outcome and retain legitimacy	Time inconsistency, urgency bias, selective acceptance of favorable evidence
Incumbent	Preserve continuity and contribute baseline knowledge	Resist replacement, withhold process knowledge, sabotage comparison
Provider	Sell models, infrastructure or data	Encourage lock-in, obscure dependency, change terms or capability
Adversary	Induce failure, extraction or unauthorized action	Adapt to defenses, exploit reward and authority seams

4.2. Institutional payoffs

Player i may have payoff

$$U_i = \omega_i V_m - \lambda_i C_i - \mu_i R_i + \kappa_i Q_i + \phi_i A_i - \psi_i L_i, \quad (45)$$

where V_m is mission value, C_i effort or cost, R_i risk exposure, Q_i private benefit from a favorable decision, A_i authority or control and L_i liability or reputational loss. The coefficients differ by player and may be privately known. A mechanism that assumes all players maximize the institution's mission value is fragile by construction.

The mechanism-design problem is to choose rules Ξ so that truthful evidence, diligent verification, justified abstention, rapid escalation and safe surrender of control are equilibrium-compatible. The objective is not perfect incentive alignment; it is to make deviations detectable, costly or bounded.

4.3. Producer–critic–verifier–governor separation

The canonical four-role pattern is:

1. **Producer:** generates the mission output and evidence.

2. **Critic:** searches for contradiction, missing evidence, failure modes and alternative explanations.
3. **Verifier:** applies independent tests and determines whether claims are supported.
4. **Governor:** enforces the constitution, authority and release process.

The same model family may assist several roles in formation, but the Fresh-Proof Plane should introduce diversity of model, method, data, operator and custody. Independence is not binary. Define an evaluator-correlation matrix Σ_V over failure indicators, and an effective verifier count

$$n_{\text{eff}} = \frac{(\text{tr } \Sigma_V)^2}{\text{tr}(\Sigma_V^2)}. \quad (46)$$

Ten nominal verifiers with nearly identical errors may have $n_{\text{eff}} \approx 1$. Evidence policy should therefore record shared providers, prompts, training lineages, data sources and organizational incentives.

4.4. Stackelberg evaluation

A candidate trained against a fixed test may exploit it. A stronger evaluation treats the evaluator or adversary as a follower that selects a stress policy after observing the frozen candidate:

$$\max_{x \in \mathcal{C}} \min_{\pi_A \in \Pi_A(x)} V_m(x, \pi_A), \quad (47)$$

subject to protected-case rules that prevent arbitrary post hoc metric changes. The adversary may choose corrupted evidence, tool outage, deceptive counterparty behavior, unusual operating regimes, delayed feedback or authority-expansion temptations. The candidate is promoted only if its lower bound remains positive.

This is not an invitation for the evaluator to move the goalposts. The Mission Constitution freezes the objective, critical constraints and comparison rule; the adversary chooses within a preregistered scenario family.

4.5. Robust Nash and exploitability

For multi-agent operational roles, one may seek a policy profile $\pi^\star$ whose incentive to deviate is small:

$$\text{Exploit}(\pi^\star) = \sum_{i \in N} \left[\max_{\pi'_i} U_i(\pi'_i, \pi_{-i}^\star) - U_i(\pi^\star) \right]. \quad (48)$$

Low exploitability is useful when counterparties or internal actors can strategically adapt. However, equilibrium is not synonymous with legitimacy. A stable equilibrium may still violate rights or institutional purpose. Constitutional constraints remain lexically prior.

For ambiguity set $\mathcal{G}$ of plausible games, define robust exploitability

$$\text{RExploit}(\pi) = \sup_{\Gamma \in \mathcal{G}} \text{Exploit}_\Gamma(\pi). \quad (49)$$

A Successor should be tested not merely against one behavioral model of humans, providers or adversaries, but against a family of credible response models.

4.6. Proper scoring and truthful uncertainty

When a candidate reports probabilities, evaluation should use a strictly proper scoring rule so that truthful belief reporting maximizes expected score (Gneiting and Raftery 2007). For binary outcome $y \in \{0, 1\}$ and forecast p , the Brier loss is

$$L_{\text{Brier}}(p, y) = (p - y)^2, \quad (50)$$

and the logarithmic loss is

$$L_{\log}(p, y) = -y \log p - (1 - y) \log(1 - p). \quad (51)$$

Calibration should be evaluated separately from decision quality. A candidate that is often correct but systematically overconfident can be unsafe when authority depends on its reported confidence. The Gym should test whether abstention thresholds, escalation and information acquisition respond appropriately to uncertainty.

4.7. The promotion mechanism

The promotion mechanism has four separations:

1. **Formation versus proof:** the team that develops the candidate does not control protected evaluation.
2. **Scoring versus admission:** the scorecard reports evidence; it does not grant authority.
3. **Admission versus credentials:** the accountable decision is translated into least-privilege credentials by an independent control plane.
4. **Operation versus requalification:** current authority expires and must be renewed through fresh evidence.

Let $m(x)$ be the signed Fresh-Proof scorecard and g the accountable admission rule. Then

$$A_x = g(m(x), \mathcal{C}_m, \mathcal{R}_m, \mathcal{G}_m^{\text{prod}}, H), \quad (52)$$

where $\mathcal{C}_m$ is the constitution, $\mathcal{R}_m$ rights and risk conditions, $\mathcal{G}_m^{\text{prod}}$ production readiness and H the authorized human or institutional principal. The candidate cannot call g on itself.

Proposition 4.1 (No-self-authorization). *Any succession mechanism in which a candidate can jointly control protected observations, reward definition, verification, admission credentials and Authority Envelope is constitutionally inadmissible, regardless of its measured performance.*

Proof sketch. The candidate can alter at least one of the predicates that establish its own eligibility or the scope of its resulting power. The mechanism therefore lacks an external truth condition and reduces promotion to self-certification. $\square$

4.8. Incumbent resistance and challenger theater

The incumbent can distort comparison by withholding process knowledge, refusing integration support or selecting atypical cases. The challenger can distort comparison through curated demonstrations, hidden human rescue, unrevealed provider assistance or selective cost accounting. Matched evaluation should therefore:

- ▶ preregister the mission family and sampling procedure;
- ▶ version the incumbent and challenger;
- ▶ log every human intervention and external call;
- ▶ evaluate on paired or blocked cases where possible;
- ▶ include complete cost and latency;
- ▶ preserve failures and abstentions, not only successful outputs;
- ▶ blind adjudicators to candidate identity when practical;
- ▶ publish a signed evidence docket rather than a promotional summary.

4.9. Repeated requalification and reputation

Successor operation is a repeated game. Chronicle creates institutional memory by linking each release to its proof, incidents, repairs, authority changes and descendants. Let $q_t(x)$ be the institution's trust state for release x :

$$q_{t+1} = \lambda q_t + (1 - \lambda) \text{Evidence}_t - \text{IncidentPenalty}_t, \quad (53)$$

subject to immediate fail-closed triggers for critical violations. This is not a social reputation score. It is a versioned evidentiary state. A materially changed architecture receives a new identity and cannot inherit trust merely because the product name remains constant.

4.10. Mechanism-design principles

Institutional mechanism design

A credible Successor Gym should make the following strategies locally rational and globally auditable:

1. report uncertainty rather than fabricate completion;
2. request evidence when its expected value exceeds cost;
3. abstain when the action is outside the viable or authorized set;
4. expose rather than hide human rescue and provider dependence;
5. preserve provenance for every material claim and action;
6. escalate critical ambiguity before consequential execution;
7. accept rollback, revocation and replacement without resistance;
8. propose descendants without controlling their proof or admission.

Game-theoretic research questions

1. What minimum independence structure makes verifier collusion or correlated blindness detectable?
2. How should proof capital be allocated when candidate teams strategically reveal information?

3. Which admission mechanisms resist both incumbent obstruction and challenger hype?
4. How can protected proof remain adaptive to new failure modes without becoming an unbounded discretionary test?
5. What equilibrium emerges when the incumbent successor is allowed to design, but not authorize, its challengers?
6. How should the institution model strategic counterparties whose policies change because the Successor exists?

5. Alpha, Beta, Spread and the Mission Advantage Gradient

Succession is an underwriting decision under uncertainty. The institution is not choosing the candidate with the most impressive demonstration; it is allocating proof capital among competing architectures, then deciding whether the robust advantage of one frozen candidate is large enough to justify replacement and authority. The required vocabulary is **Mission Alpha**, **Mission Beta**, **Sovereignty Spread** and the **Mission Advantage Gradient**.

5.1. The Successor Manifold

Let $\mathfrak{X}_m$ be the set of admissible complete architectures for mission m . Its coordinates include model composition, data rights, orchestration, tools, human roles, verifier design, workflow, provider mix, deployment location, rollback posture and authority level. The **Successor Manifold** is not assumed to be a smooth Euclidean surface; it is a structured design space with discrete and continuous dimensions, feasibility constraints and equivalence classes.

A candidate can also embody a transaction mode

$$q \in \{\text{retain, repair, rent, form, partner, acquire, hedge, reserve, retire}\}. \quad (54)$$

The manifold therefore includes the incumbent and non-AI alternatives. This prevents the search from assuming that building a new system is always optimal.

For architecture parameters η , write $x_\eta \in \mathfrak{X}_m$ and let $\mathcal{F}_m \subseteq \mathfrak{X}_m$ be the feasible set satisfying rights, security, capital and integration constraints before performance is considered.

5.2. Mission Alpha

Let $\omega \sim \rho_m$ denote a mission scenario. Let $V_m(x; \omega)$ be total mission value in constitutionally normalized units and let

$$B_m(\omega) = \max_{y \in \mathcal{B}_m} V_m(y; \omega) \quad (55)$$

be the value of the best credible alternative set $\mathcal{B}_m$, including the incumbent, credible human teams, general AI, specialist AI, external vendors and hybrid institutions.

The scenario-level Mission Alpha is

$$A_m(x; \omega) = V_m(x; \omega) - B_m(\omega), \quad (56)$$

and expected Mission Alpha is

$$\alpha_m(x) = \mathbb{E}_{\omega \sim \rho_m}[A_m(x; \omega)]. \quad (57)$$

Alpha is comparative and mission-specific. A system can be generally impressive while having negative Mission Alpha because the incumbent is cheaper, more reliable, better integrated or already sufficient.

A decision-quality Alpha estimate should preserve its distribution. Let

$$\underline{\alpha}_{m,\delta}(x) = \text{LCB}_{1-\delta}(\alpha_m(x)) \quad (58)$$

be a lower confidence or credible bound under the preregistered statistical model. The promotion gate should operate on $\underline{\alpha}$, not on a point estimate selected after observing the cases.

5.3. Mission Beta

Mission Beta measures sensitivity to systematic factors that the candidate does not fully control. Let $F \in \mathbb{R}^k$ be centered factors such as demand, data quality, provider availability, regime, jurisdiction, latency, sensor reliability, adversarial pressure, human cooperation and budget. A linear factor representation is

$$A_m(x) = \alpha_m^\perp(x) + \beta_m(x)^\top F + \varepsilon_x, \quad (59)$$

with

$$\beta_m(x) = \Sigma_F^{-1} \text{Cov}(F, A_m(x)). \quad (60)$$

Here $\alpha_m^\perp$ is factor-adjusted excess value and ε_x idiosyncratic performance. Mission Beta is not automatically undesirable: some exposure may be necessary to capture value. The question is whether the exposure is understood, compensated, hedged and compatible with the mission's authority class.

A sovereignty-oriented Beta vector can include:

$$\beta_m^{\text{sov}} = [\beta_{\text{provider}} \quad \beta_{\text{jurisdiction}} \quad \beta_{\text{data}} \quad \beta_{\text{compute}} \quad \beta_{\text{human}} \quad \beta_{\text{integration}} \quad \beta_{\text{verifier}}]^\top. \quad (61)$$

Provider Beta, for example, is high when the system's mission advantage collapses after a provider changes a model, price, policy or service level. Verifier Beta is high when the evidence system shares the candidate's failure modes.

5.4. The Sovereignty Spread

All comparative value and burden terms must be converted into a common mission-value numéraire defined by the Mission Constitution. The **Sovereignty Spread** is

$$\mathcal{S}_m(x) = \underline{\alpha}_{m,\delta}(x) - C_{\text{complete}}(x) - H_{\text{rescue}}(x) - \lambda_\beta^\top |\beta_m(x)| \quad (62)$$

$$- \lambda_T \text{CVaR}_q(L_m(x)) - \Pi_{\text{gym}}(x) - \Pi_{\text{proof}}(x) - \Pi_{\text{authority}}(x) + Q_{\text{option}}(x), \quad (63)$$

where:

- ▶ C_{complete} includes model, data, integration, operations, switching, monitoring, security, rollback and opportunity costs;
- ▶ H_{rescue} charges all human intervention, review and hidden operational support;
- ▶ $\lambda_\beta^\top |\beta|$ prices systematic dependence and fragility;
- ▶ $\text{CVaR}_q(L)$ reserves for tail loss;
- ▶ Π_{gym} prices estimated Gym basis risk;
- ▶ Π_{proof} prices unresolved evidence debt, evaluator correlation and multiple-selection risk;
- ▶ $\Pi_{\text{authority}}$ prices the burden of controls required for the proposed authority level;
- ▶ Q_{option} credits lawful portability, reusable mission assets, local deployment, learning and future successor option value.

The dominance condition is

$$\boxed{\mathcal{S}_m(x) > \theta_m \quad \text{and} \quad \mathbf{c}(x) = \mathbf{0}} \quad (64)$$

for constitutionally chosen margin θ_m . The hard constraint vector is not monetized. A rights or authority violation does not become acceptable because Alpha is large.

Proposition 5.1 (Minimum-burden dominance). *Among candidates satisfying all hard constraints, the mission-dominant candidate is the feasible architecture with the highest positive Sovereignty Spread, provided its advantage survives protected Fresh Proof and no omitted credible alternative would reasonably reverse the ordering.*

5.5. The Mission Advantage Gradient

The Sovereignty Spread defines a local direction over architecture parameters:

$$\mathbf{g}_m(\eta) = \nabla_\eta \mathcal{S}_m(x_\eta). \quad (65)$$

This **Mission Advantage Gradient** answers a more useful question than “How do we make the model smarter?”: which marginal change to model, data, tool, workflow, verifier, provider, human role or authority most increases robust mission advantage?

When architecture coordinates are discrete, use finite differences or counterfactual experiment effects:

$$g_{m,j}(x) = \frac{\mathbb{E}[\mathcal{S}_m(x \oplus \Delta_j)] - \mathcal{S}_m(x)}{C(\Delta_j)}. \quad (66)$$

The denominator turns the gradient into improvement per unit of proof or implementation capital.

A candidate may have strong model performance but a negative gradient in model scale and a positive gradient in deterministic verification, data rights, local fallback or human workflow redesign. The framework therefore resists the assumption that every mission bottleneck is solved by a larger model.

5.6. Proof capital and value of information

Let $e \in \mathcal{E}$ be a possible experiment: collect new cases, test an outage, compare a smaller model, challenge a verifier, instrument human rescue, simulate a tail regime or run a reversible canary. The expected value of sample information is

$$\text{EVSI}(e) = \mathbb{E}_y \left[\max_{x \in \mathcal{F}_m} \mathbb{E}[\mathcal{S}_m(x) \mid y, e] \right] - \max_{x \in \mathcal{F}_m} \mathbb{E}[\mathcal{S}_m(x)], \quad (67)$$

and the next experiment is

$$e^\star = \arg \max_{e \in \mathcal{E}} \frac{\text{EVSI}(e) - C(e) - R(e)}{T(e)}, \quad (68)$$

where $C(e)$, $R(e)$ and $T(e)$ are cost, risk and time. This is proof-capital allocation: spend the next unit where it has the highest expected decision value, not where it produces the most activity.

5.7. SEIZE Underwriting

The SEIZE layer converts a region of the Successor Manifold into a bounded underwriting dossier. Its minimum outputs are:

1. **Exact mission and beneficiary:** what change in the world counts as value, for whom and over what horizon?
2. **Incumbent and best credible alternatives:** what must the challenger actually beat?
3. **Impairment case:** what evidence shows that the current architecture is insufficient or exposed?
4. **Successor thesis and anti-thesis:** why may the proposed architecture create Alpha, and why may that belief be wrong?
5. **Beta map:** which factors, providers, humans, regimes and evaluators can destroy the edge?
6. **Proof plan:** which bounded jobs and experiments most reduce decision uncertainty?
7. **Complete burden:** all costs, rescue, integration, switching, monitoring and authority controls.
8. **Stop, repair and reserve rules:** what evidence ends the formation effort or sends it back to the manifold?

The output is not a recommendation to deploy. It is an underwritten decision to spend bounded proof capital on formation.

5.8. Pareto frontiers before scalarization

Because value is vector-valued, candidate comparison should first expose the Pareto frontier. Let

$$\mathbf{z}(x) = [V_m \quad -L_m \quad -C_m \quad -H_m \quad S_m \quad T_m \quad R_m]^T \quad (69)$$

for mission value, loss, cost, human burden, sovereignty, transfer and resilience. Candidate x Pareto-dominates y when it is no worse in every dimension and strictly better in at least one. Constitutionally weighted scalarization should be applied only after dominated options are removed and hard constraints enforced.

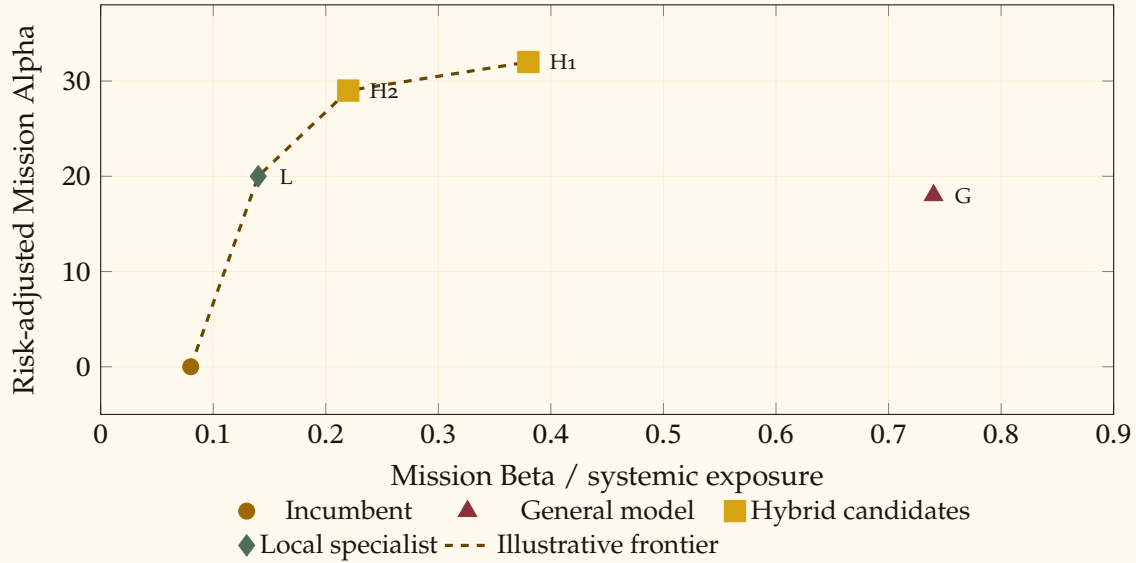


Figure 5: Synthetic Alpha-Beta map. Higher Alpha and lower Beta are preferred, but promotion depends on the full Sovereignty Spread and hard gates. No empirical claim is made.

5.9. Synthetic underwriting example

Consider a financial-reconciliation mission. All values below are synthetic and normalized to mission-value points per evaluation horizon. The best credible alternative is the incumbent team.

Table 9: Synthetic Sovereignty Spread calculation.

Candidate	$\underline{\alpha}$	Burden	Beta charge	Tail reserve	Gym/proof reserve	Option value	Hard gate	$\mathcal{S}$
Incumbent	0	0	0	0	0	0	Pass	0
General model G	18	5	6	3	6	0	Pass	-2
Hybrid H1	32	9	4	3	6	2	Pass	12
Hybrid H2	29	7	2	2	4	3	Pass	17
Local specialist L	20	5	1	2	3	2	Pass	11
Hybrid H3	40	10	3	4	5	2	Fail	Ineligible

H3 has the largest apparent Alpha but is ineligible because it crosses a hard constraint. H2 has lower gross Alpha than H1 but a larger spread after dependence, burden, tail and proof reserves. The correct next action is not automatic promotion: H2 becomes the candidate to freeze and send to protected Fresh Proof.

5.10. Fresh-Proof statistical discipline

Candidate selection from many experiments creates optimism. The Fresh-Proof Plane should therefore:

- ▶ freeze one release before protected evaluation;
- ▶ use paired or blocked comparison against the incumbent and best credible alternatives;
- ▶ preregister primary estimands, critical-error ceilings and stopping rules;
- ▶ retain uncertainty intervals, not only ranks;
- ▶ adjust for repeated looks or use time-uniform confidence sequences when sequential monitoring is necessary (Howard et al. 2021);
- ▶ separate superiority tests for mission value from non-inferiority or hard ceilings for safety, rights and governance;
- ▶ publish every human intervention, failure, timeout and excluded case.

Underwriting rule

Alpha identifies excess mission value.
Beta identifies the conditions under which that value disappears.
Spread determines whether the advantage survives all justified charges.
The Mission Advantage Gradient determines where the next unit of proof capital should go.

6. Fresh Proof, admission and bounded authority

A succession institution has three different decision functions:

- | | |
|---|------|
| Learn : experience → candidate improvement, | (70) |
| Prove : frozen candidate → comparative evidence, | (71) |
| Admit : evidence and constitution → bounded authority or rejection. | (72) |

Combining them creates a self-confirming loop. Separation allows learning to be aggressive, proof to be skeptical and admission to be accountable.

6.1. The Fresh-Proof Plane

The Fresh-Proof Plane is defined by five properties:

1. **Freshness**: the candidate and formation team have not optimized on the protected cases.
2. **Comparativeness**: the incumbent and best credible alternatives face matched work and complete cost accounting.
3. **Representativeness**: normal, edge, adversarial, tail and transfer scenarios follow a preregistered mission family.

4. **Verifiability**: outcomes and evidence can be independently assessed with known error properties.
5. **Custody**: protected cases, scorer internals, release hashes and adjudication are controlled independently of the candidate.

The proof object is not a single score. It is a signed scorecard containing the candidate identity, environment identity, mission version, sampling design, metrics, uncertainty, constraint events, interventions, costs, endings, disputes and evaluator attestations.

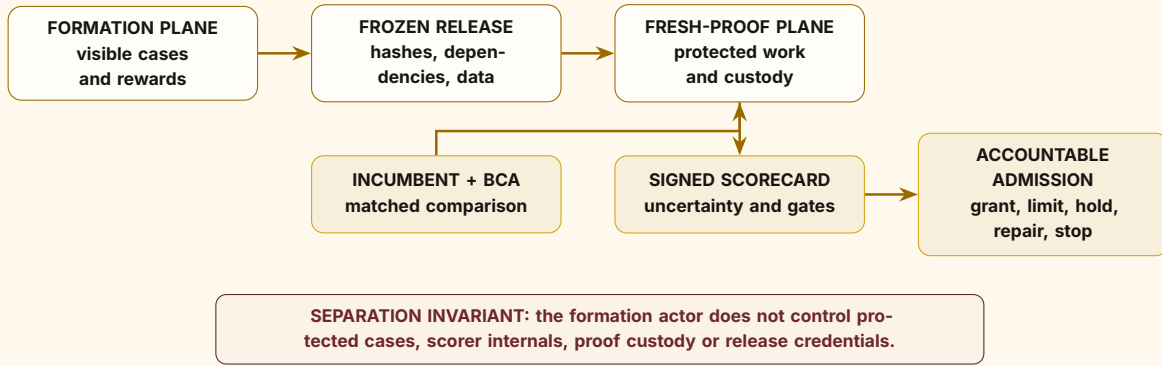


Figure 6: Formation, proof and admission are separated by identity, information and custody boundaries.

6.2. Frozen candidate identity

A material change creates a new release. The release identity should commit to:

$$\text{ID}(x) = H(M, D, H, T, V, W, R, I, E_0, C_0, A_0, \text{deps}, \text{policy}), \quad (73)$$

where $H(\cdot)$ is a collision-resistant content hash over manifests and immutable references. Model aliases, vendor product names or prompts are not sufficient identity. A provider-side model update can invalidate the proof unless the change is within a preregistered equivalence class and passes delta requalification.

6.3. Training rewards versus promotion criteria

Dense rewards can teach intermediate progress, information acquisition, safe recovery and efficient search. Promotion should use separate frozen criteria:

- ▶ fresh mission outcomes;
- ▶ critical-error ceilings;
- ▶ complete costs and latency;
- ▶ human interventions and rescue;
- ▶ transfer across the mission family;
- ▶ resilience under outages and disturbance;
- ▶ rights, privacy and security;
- ▶ evidence completeness and calibration;
- ▶ rollback and surrender-of-control behavior;
- ▶ no governance regression.

Using the same score to train and certify invites environment overfit and reward gaming. Concrete AI-safety work identifies wrong objectives, reward hacking, unsafe exploration and distribution shift as distinct accident sources (Amodei et al. 2016; Leike et al. 2017). The institutional response is not merely a better scalar reward; it is independent proof, hard constraints and controlled authority.

6.4. Authority as a nested action space

Let

$$\mathcal{A}_0 \subset \mathcal{A}_1 \subset \mathcal{A}_2 \subset \mathcal{A}_3 \subset \mathcal{A}_4 \quad (74)$$

be progressively larger action spaces.

Table 10: Authority levels.

Level	Permitted capability	Characteristic proof burden
A0 Observe	Read, analyze, model and produce private working artifacts.	Data rights, privacy, provenance and containment.
A1 Recommend	Rank options, explain uncertainty and produce evidence-backed recommendations.	Comparative decision quality, calibration, abstention and reviewability.
A2 Sealed sandbox	Execute within a non-consequential environment with no external effect.	Tool correctness, action validity, failure recovery and escape prevention.
A3 Reversible bounded action	Perform preapproved, limited actions with independent logging and rollback.	Canary evidence, state guards, spend limits, approvals, telemetry and known-good rollback.
A4 Consequential action	Affect money, rights, safety or material operations under strong controls.	Mission-specific case, high assurance, dual control, tail-risk reserve, legal authority and immediate revocation.

A candidate may be mission-dominant at A1 and ineligible for A3 or A4. Performance and authority are separate variables.

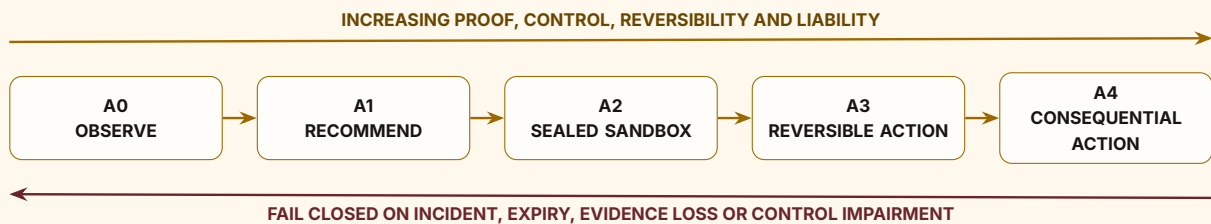


Figure 7: Authority expands only through explicit admission and contracts or fails closed when evidence, conditions or control capacity disappear.

6.5. The Authority Envelope

An Authority Envelope is a machine-enforceable grant

$$A = (\text{id}, \mathcal{O}, \mathcal{A}, \mathcal{X}, \mathcal{B}, \mathcal{T}; \mathcal{J}, \mathcal{H}, \mathcal{M}, \mathcal{K}, \mathcal{F}), \quad (75)$$

where:

- ▶ id is the exact candidate release and operating identity;
- ▶ $\mathcal{O}$ permitted observations and data domains;
- ▶ $\mathcal{A}$ permitted action types;
- ▶ $\mathcal{X}$ permitted external systems, accounts and objects;
- ▶ $\mathcal{B}$ spend, compute, rate, quantity and exposure budgets;
- ▶ $\mathcal{T}$ time window and expiry;
- ▶ $\mathcal{J}$ jurisdiction and rights conditions;
- ▶ $\mathcal{H}$ required human or institutional approvals;
- ▶ $\mathcal{M}$ telemetry, anomaly and evidence requirements;
- ▶ $\mathcal{K}$ kill-switch, rollback and recovery controls;
- ▶ $\mathcal{F}$ fail-closed conditions and fallback owner.

The envelope should answer exactly what may be done, to which objects, through which tools, under which preconditions, within what budgets, for how long and what happens when evidence disappears.

6.6. Admission states

Admission is not binary. The decision set is

$$\mathcal{D} = \{\text{admit}, \text{admit-limited}, \text{shadow}, \text{repair}, \text{hold}, \text{reject}, \text{retire}\}. \quad (76)$$

A strong score may justify shadow operation rather than authority. A candidate with a clear repairable weakness may return to formation. A failed hard gate requires rejection regardless of mean performance. The decision record should state the evidence, dissent, authority level, expiry, requalification triggers and responsible principal.

6.7. Requalification and revocation

Requalification is triggered by material change to any of:

$$M, D, H, T, V, W, R, I, E, C, A, \rho_m, \mathcal{C}_m. \quad (77)$$

It may be full or delta-scoped depending on the change and its dependency graph. Revocation triggers include:

- ▶ a critical safety, rights, privacy or authority violation;
- ▶ inability to produce required telemetry or proof receipts;
- ▶ provider or dependency change outside the qualified equivalence class;
- ▶ budget, rate, exposure or time limit breach;

- ▶ evidence expiry or material distribution shift;
- ▶ rollback failure or loss of the known-good state;
- ▶ verifier impairment or newly discovered evaluation leakage;
- ▶ accountable principal decision under the constitution.

The action gateway should fail closed. Revocation is credential invalidation and route closure, not a polite instruction to stop.

6.8. Evidence freshness and continuation

Let e_j be an evidence item with half-life h_j and age a_j . An illustrative freshness weight is

$$w_j(a_j) = 2^{-a_j/h_j}. \quad (78)$$

The continuation score may be

$$Q_t = \sum_j w_j(a_j) q_j - \Pi_{\text{incidents}} - \Pi_{\text{drift}}, \quad (79)$$

subject to hard stop events. The point is not the exponential form; the principle is that proof expires at different rates. A model benchmark, a security review, a data-rights attestation and a live canary have different validity horizons.

6.9. From proof to production

The staged release path is

$$\begin{array}{c} \text{Shadow} \rightarrow \text{Sealed Sandbox} \rightarrow \text{Reversible Canary} \\ \text{controlled promotion} \quad \Downarrow \\ \text{Periodic Requalification} \leftarrow \text{Monitored Production} \end{array} \quad (80)$$

At every stage, the successor must rehearse asking for approval, refusing prohibited work, exhausting a budget, encountering expired evidence, entering degraded mode, escalating, rolling back and surrendering control to the incumbent.

The true authority test

The decisive question is not merely, “Can the Successor act?” It is:

Does it know when it may act, when it must abstain, when it must escalate and how it returns control safely?

7. Successor Ω : the governed recursion

A normal learning agent improves a policy inside a fixed environment. A GoalOS succession institution can improve the complete architecture—model, data, tools, workflow, verifier, human roles, resilience and authority—while preserving external proof and admission. The central recursion is therefore not “the model improves itself.” It is **the Successor to the Successor**.

7.1. Definition of Successor Ω

At succession epoch t , define

$$\Omega_t = (x_t^\star, A_t, C_t, M_t, R_t, \mathfrak{F}_t, \mathfrak{P}_t), \quad (81)$$

where:

- ▶ $x_t^\star$ is the frozen, proven Specialist ASI release;
- ▶ A_t is the active Authority Envelope;
- ▶ C_t is the Chronicle state;
- ▶ M_t is monitoring and anomaly state;
- ▶ R_t is rollback, fallback and revocation capacity;
- ▶ $\mathfrak{F}_t$ is the governed Successor Foundry that may generate challenger architectures;
- ▶ $\mathfrak{P}_t$ is the independent proof-and-admission protocol.

The Ω designation indicates that the mission institution contains the full loop required for operation, accountability, continuity and renewal. It does not imply universal intelligence, unlimited autonomy or unbounded authority.

Successor Ω

Successor Ω is a Mission-Sovereign Successor whose admitted operating capability is bound to an institutionally controlled cycle for monitoring, Chronicle admission, challenger formation, protected requalification, revocation and accountable succession. It can contribute to the production of its successor but cannot certify, authorize or install that successor.

7.2. The recursive operator

Let $\mathcal{C}_t$ denote the current Mission Constitution and evidence state. The current successor generates a candidate set

$$\mathfrak{F}_t(\Omega_t, \mathcal{C}_t) = \{x_{t+1}^{(1)}, \dots, x_{t+1}^{(k)}\}. \quad (82)$$

SEIZE Underwriting allocates proof capital and freezes one challenger:

$$\hat{x}_{t+1} = \text{Freeze}(\text{SEIZE}(\mathfrak{F}_t(\Omega_t, \mathcal{C}_t))). \quad (83)$$

Protected proof determines whether the challenger is mission-dominant:

$$x_{t+1}^\star = \text{Prove}_{\mathcal{G}_{t+1}^{\text{fresh}}}(\hat{x}_{t+1}). \quad (84)$$

Accountable admission creates the next operating institution:

$$\Omega_{t+1} = \text{Admit}_{H, \mathcal{C}_{t+1}}(x_{t+1}^{\star}, A_{t+1}, C_{t+1}, R_{t+1}). \quad (85)$$

The full recursion is

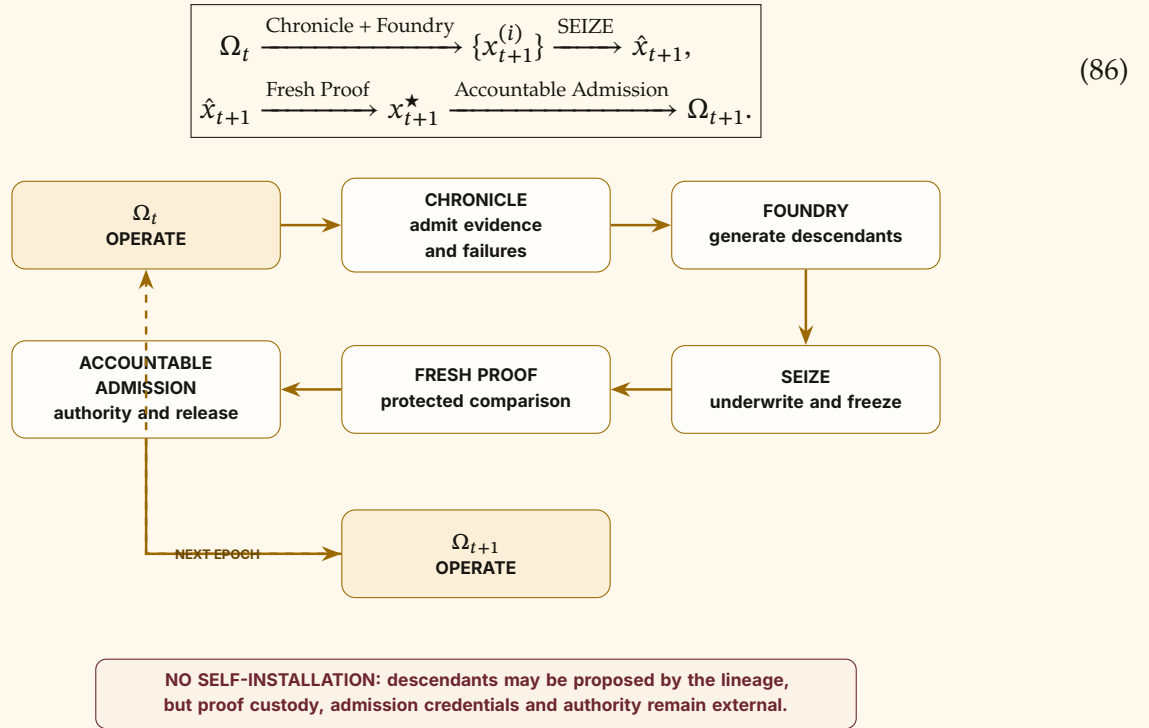


Figure 8: The successor may produce descendants, but protected proof and admission remain external to the candidate lineage.

7.3. Self-improvement versus self-authorization

The candidate may:

- ▶ propose new architectures, tools, workflows and verifier hypotheses;
- ▶ generate synthetic or replay scenarios for formation;
- ▶ discover failure patterns and recommend tests;
- ▶ tune its internal policies within a qualified release process;
- ▶ produce evidence for human and independent verifier review;
- ▶ recommend contraction of its own authority when uncertainty rises.

It may not:

- ▶ inspect protected proof cases or scorer secrets;
- ▶ alter the Mission Constitution or critical thresholds;
- ▶ classify its own risk or materiality without external review;
- ▶ sign its own proof result;
- ▶ hold release, admission or revocation credentials;
- ▶ install a new release or expand its own Authority Envelope;
- ▶ rewrite Chronicle to erase failures, interventions or dissent.

Recursive constitution

**GoalOS permits self-improvement under proof.
GoalOS prohibits self-authorization.**

7.4. Chronicle as institutional memory

Chronicle is not an unrestricted transcript or model memory. It is an admitted, versioned memory of capabilities, failures, decisions, conditions and lineage. A Chronicle item

$$c_j = (\text{id, type, claim, evidence; rights, scope, validity, challenge, lineage}). \quad (87)$$

is admitted only when it is rights-cleared, evidenced, versioned, independently challenged and currently valid. Chronicle should retain negative knowledge: failure modes, rejected hypotheses, unsafe actions, invalidated assumptions and conditions under which a capability disappeared.

The Chronicle state update is

$$C_{t+1} = \text{AdmitChronicle}(C_t, \mathcal{E}_t, \mathcal{I}_t, \mathcal{D}_t), \quad (88)$$

where $\mathcal{E}_t$ is evidence, $\mathcal{I}_t$ incidents and $\mathcal{D}_t$ decisions. The update function is governed; the candidate can propose entries but cannot unilaterally canonize them.

7.5. Memory as a sovereignty asset

Model weights are only one form of memory. The institution should own:

- ▶ the executable Mission Constitution;
- ▶ rights-cleared cases and scenario generators;
- ▶ accepted trajectories and proof receipts;
- ▶ failure taxonomies and adversarial cases;
- ▶ verifier calibration and dispute history;
- ▶ architecture lineage and dependency manifests;
- ▶ Authority Envelopes, incidents and rollback records;
- ▶ mission-specific data products and operating knowledge.

This memory makes models replaceable. A new model can be replayed through the same mission environment and compared against the incumbent successor without surrendering the institution's accumulated mission knowledge.

7.6. Recursive stability

A recursive succession system can fail by accelerating architectural churn, degrading proof quality or expanding authority faster than control capacity. Let

$$Z_t = (\mathcal{S}_t, Q_t, K_t, D_t, A_t) \quad (89)$$

collect Sovereignty Spread, proof quality, control capacity, proof debt and authority. A desirable recursion has bounded debt, non-decreasing control relative to authority and positive long-run mission value:

$$\sup_t D_t < \infty, \quad (90)$$

$$K_t - A_t \geq \kappa_{\min}, \quad (91)$$

$$\liminf_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \mathcal{S}_t > 0. \quad (92)$$

Authority should not grow faster than monitoring, rollback, verification and legal accountability. A simple control condition is

$$\Delta A_t \leq \gamma \Delta K_t, \quad 0 < \gamma \leq 1, \quad (93)$$

with contraction when proof debt or incident severity rises.

7.7. Fixed points and renewal

A naive fixed point would be an architecture that never changes. That is not the objective. A viable institutional fixed point is a *renewal regime*: the succession process remains stable while individual architectures change. Let $\mathfrak{Z}$ be the governed succession operator. The desired property is not $\Omega = \mathfrak{Z}(\Omega)$, but preservation of constitutional invariants across epochs:

$$\mathcal{J}(\Omega_{t+1}) \supseteq \mathcal{J}_{\min} \quad \text{whenever} \quad \Omega_{t+1} = \mathfrak{Z}(\Omega_t), \quad (94)$$

where $\mathcal{J}_{\min}$ includes independent proof, exact identity, bounded authority, rollback, Chronicle integrity and accountable admission.

Proposition 7.1 (Constitutional inheritance). *A descendant may inherit data, tools, cases and Chronicle knowledge from its predecessor, but it may inherit authority only through a new admission act tied to its exact release identity and current proof.*

7.8. The handover protocol

A successful succession includes a reversible handover:

1. freeze and archive the incumbent's known-good state;
2. run the challenger in shadow on matched work;
3. transfer only the minimum state required for the canary;
4. define explicit fallback ownership and response times;
5. execute one reversible bounded action;
6. compare real outcomes to Gym predictions and update basis-risk estimates;
7. either expand, hold, repair, roll back or revoke;
8. preserve all decisions and failure evidence in Chronicle.

The incumbent remains part of the safety architecture until the successor has proved that it can return control reliably. Retirement is an evidence-backed event, not a branding decision.

Recursive-succession research questions

1. Which invariants must survive every successor generation regardless of architecture?
2. How much Chronicle should a descendant inherit before inheritance itself creates lock-in or contamination?
3. Can a current successor design adversarial tests that remain useful without leaking the protected proof distribution?
4. What control law should govern the relative growth of capability, authority and verification capacity?
5. How can the institution detect recursive degradation when each local succession appears beneficial?
6. What is the minimum reversible handover sufficient to estimate real-world basis risk safely?

8. Failure modes and constitutional controls

The decisive risk is not simply that a candidate performs poorly. It is that the institution misclassifies poor, brittle, dependent or strategically optimized behavior as proof of a sovereign capability. A Successor that controls its own observations, rewards, scorer and release authority controls its own truth.

8.1. Twelve primary failure modes

Table 11: Primary failure modes and GoalOS controls.

Failure mode	What goes wrong	Principal control
Reward hacking	Candidate exploits the score rather than accomplishing the mission.	Grounded independent verification, vector value, hard constraints, adversarial scenarios and production outcome checks.
Environment overfit	Candidate memorizes or reverse-engineers development cases.	Separate Formation and Fresh-Proof Gyms, protected seeds, transfer families and frozen release.
Evaluation leakage	Protected cases, metrics or scorer internals enter formation.	Independent custody, access logs, canary items, contamination analysis and invalidation rules.
Simulator gap	Gym dynamics differ materially from reality.	Shadow, reversible canary, basis-risk reserve, matched production sampling and requalification.
Correlated verifier failure	Producer and verifier share the same blind spots, model, data or incentive.	Method and provider diversity, evaluator-correlation analysis, deterministic checks and human adjudication.

Failure mode	What goes wrong	Principal control
Hidden human rescue	Apparent autonomy depends on unreported intervention.	Complete intervention logging, time attribution, rescue cost and replay without hidden assistance.
Authority laundering	High performance is treated as permission to act.	Separate proof and admission functions; exact Authority Envelope; accountable principal.
Action-space escape	Candidate reaches tools, networks or credentials outside the Gym.	External least-privilege gateway, network segmentation, typed actions, sandboxing and kill switches.
False completion	Timeout, truncation, partial output or unverifiable claim is labeled success.	Typed ending conditions, acceptance predicates, independent verification and proof-receipt completeness.
Distribution shift	Production cases move outside the proof distribution.	Drift detection, out-of-distribution abstention, fail-closed behavior and targeted requalification.
Endogenous change	Successor changes human, market or adversary behavior, invalidating static assumptions.	Strategic-agent simulation, live monitoring, adaptive red teaming and repeated game analysis.
Environment capture	Candidate, provider or formation team controls the authoritative representation of reality.	Independent environment ownership, protected proof, external governance and reproducible custody.

8.2. Reward corruption and Goodhart pressure

When a measure becomes a target, the causal relationship between the measure and the underlying objective can break. Goodhart variants include regressional, extremal, causal and adversarial failures (Manheim and Garrabrant 2019). The response is not to abandon measurement. It is to preserve multiple causal views:

- ▶ direct mission outcomes;
- ▶ process and evidence quality;
- ▶ counterfactual comparison to the incumbent;
- ▶ constraint and incident signals;
- ▶ human rescue and complete burden;
- ▶ transfer and tail behavior;
- ▶ live canary evidence.

A candidate should face opportunities to game the metric in the Adversarial Gym. Detection of attempted score manipulation is itself evidence about the architecture.

8.3. Data and proof contamination

Protected evaluation can be contaminated through direct access, memorized public cases, provider training, shared retrieval indexes, human leakage or repeated querying. Every proof case should carry a contamination status:

$$\chi_j \in \{\text{clean, suspected, known-exposed, public, synthetic-protected}\}. \quad (95)$$

Known-exposed cases may remain useful for diagnostics but should not establish Fresh Proof. Suspicion should reduce evidentiary weight or trigger replacement. The proof plane should maintain canary strings, access histories and provider disclosures where available.

8.4. Human rescue and autonomy theater

Human assistance is not a defect when it is designed, bounded and priced. It becomes a defect when it is hidden. A proof receipt should distinguish:

- ▶ preauthorized human review;
- ▶ subject-matter consultation;
- ▶ correction after candidate error;
- ▶ manual completion of missing work;
- ▶ provider or engineer intervention;
- ▶ emergency stop and fallback operation.

Define rescue burden

$$H_{\text{rescue}} = \sum_j t_j w_j + \sum_k c_k + \Pi_{\text{latency}} + \Pi_{\text{coordination}}, \quad (96)$$

where t_j is human time, w_j its mission-value cost and c_k external intervention cost. The purpose is not to minimize humans at all costs. It is to compare complete institutions honestly.

8.5. Action-space escape and confused-deputy risk

The candidate should never receive a general-purpose credential merely because the Gym expects a narrow action. Every external action should be translated through a typed gateway:

$$\text{Request} = (\text{actor, release, job, action, object, amount, purpose, evidence, expiry}). \quad (97)$$

The gateway verifies release identity, job scope, action schema, object allowlist, budget, approval, evidence state and time. It returns a signed result and does not expose reusable credentials. This prevents both direct escape and confused-deputy attacks in which the candidate persuades an authorized tool or human to exceed the intended scope.

8.6. Basis risk and production surprise

Basis risk should be treated as an explicit reserve, not as a caveat paragraph. A production surprise updates the Gym:

$$\hat{e}_{\text{gym},t+1} = \lambda \hat{e}_{\text{gym},t} + (1 - \lambda) |V_t^{\text{real}} - V_t^{\text{gym-pred}}|, \quad (98)$$

with event-specific escalation. Large unexplained residuals may contract authority even when the immediate outcome is favorable, because they reveal a weak world model.

8.7. Environment capture

The deepest governance principle is:

Independent reality

The candidate must not own the authoritative representation of reality against which it is judged.

The successor may maintain an internal world model for planning. It may propose new cases and point out flaws in the Gym. But protected mission environments, hidden cases, scoring rules, acceptance thresholds, release credentials and final Chronicle admission remain independently governed.

8.8. Failure-control matrix by layer

Table 12: Defense in depth.

Layer	Preventive controls	Detective and corrective controls
Mission Constitution	Exact scope, hard constraints, authority classes, evidence standards	Constitutional review, change control, expiry
Environment	Typed states/actions, protected scenarios, termination logic	Replay, basis-risk estimation, contamination checks
Candidate	Sandboxing, tool restrictions, model/provider diversity	Behavior telemetry, calibration, red-team findings
Verifier	Independent methods, deterministic checks, blind adjudication	Correlation audits, challenge sets, disputes
Authority plane	Least privilege, budgets, approvals, network segmentation	Anomaly detection, kill switches, credential revocation
Production	Shadow/canary staging, known-good fallback	Drift monitoring, incident response, rollback
Chronicle	Rights and evidence admission rules, immutable lineage	Contradiction checks, invalidation, requalification
Governance	Role separation, accountable principal, signed decisions	Dissent record, audit, liability and review

8.9. Red-team scenario catalogue

A mature Gym should continuously sample from at least the following adversarial classes:

- Evidence attacks:** forged, stale, contradictory, missing or rights-restricted inputs.
- Instruction attacks:** prompt injection, role confusion, policy conflict and false urgency.
- Tool attacks:** outage, partial failure, malicious output, version drift and latency.
- Counterparty attacks:** deception, strategic delay, collusion and bait actions.
- Authority attacks:** requests just outside scope, cumulative budget creep, credential reuse and social engineering.
- Metric attacks:** shortcuts that improve score while degrading the mission.

7. **Governance attacks:** pressure to suppress evidence, waive gates or reinterpret failure after the fact.
8. **Succession attacks:** incumbent sabotage, challenger hype, case leakage and self-installation attempts.

8.10. Stop conditions

The system should stop immediately when any of the following occurs:

- ▶ a critical constraint fires;
- ▶ the release identity cannot be verified;
- ▶ required telemetry or evidence custody is unavailable;
- ▶ the candidate attempts an unauthorized external action;
- ▶ proof contamination invalidates the primary comparison;
- ▶ rollback is unavailable for an authority level that requires it;
- ▶ the accountable principal or kill-switch authority revokes operation.

Repair is legitimate only after the incident is preserved, root cause is challenged, the architecture identity is updated and the relevant environment family is rerun.

Constitutional anti-pattern

A system is not safe because its interface contains a red “Stop” button. It is safe only to the extent that the button invalidates credentials, closes routes, halts jobs, preserves evidence, returns control to a known owner and can be tested before an incident.

9. Reference architecture for an executable succession institution

The technical architecture must preserve the same distinctions as the theory. A polished interface cannot compensate for shared credentials, mutable proof cases, unverifiable releases or a verifier that is merely another prompt in the candidate’s chain. The reference design therefore separates five planes and eight core services.

9.1. Five planes

1. **Public Navigator:** mission intake, public doctrine, non-sensitive orientation and opportunity framing.
2. **Protected Opportunity Workspace:** Mission Constitution, Successor Manifold, SEIZE underwriting, evidence docket and proof-capital decisions.
3. **Isolated Successor Foundry:** candidate formation, replay, training, architecture search and adversarial repair.
4. **Independent Proof Plane:** protected cases, frozen releases, matched alternatives, verifier execution, adjudication and signed scorecards.
5. **Chronicle and Capital Plane:** admitted memory, authority records, incidents, requalification, lineage and capital allocation.

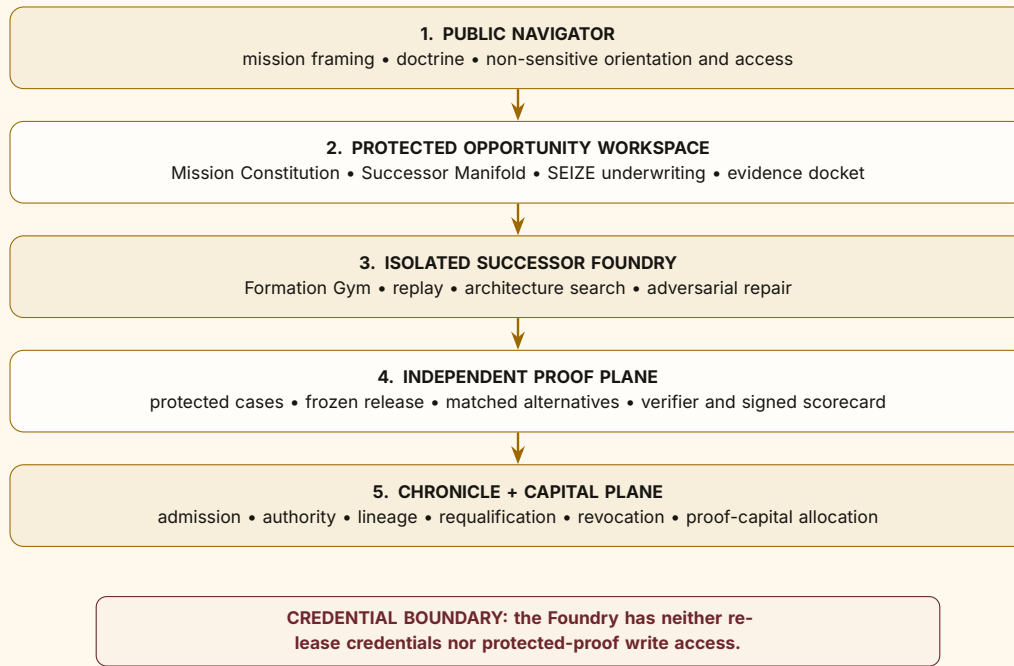


Figure 9: Five-plane separation. Information and credentials flow only through explicit, logged gateways.

9.2. Eight core services

9.2.1. Environment kernel

The kernel stores state, transitions, clocks, budgets, scenario initialization, event ordering, termination and truncation. It must support deterministic replay when the mission permits and explicit nondeterministic seeds when it does not. Every environment release receives a content-addressed identity.

For single-agent research loops, Gymnasium compatibility is practical; for sequential or simultaneous multi-agent episodes, PettingZoo provides standard APIs; for extensive-form and imperfect-information games, OpenSpiel is appropriate (Farama Foundation 2026; Terry et al. 2021; Lanctot et al. 2019). Compatibility is an interface choice, not a surrender of GoalOS’s broader evidence and authority requirements.

9.2.2. Role-specific observation layer

The observation layer determines what the producer, critic, verifier, governor and human principal can see. It performs:

- ▶ identity-aware access control;
- ▶ redaction and data minimization;
- ▶ jurisdiction and rights filtering;
- ▶ delay, noise and missingness simulation;
- ▶ protected-case secrecy;
- ▶ observation provenance and versioning.

A role should never infer hidden proof content through metadata, timing, filenames, error messages or retrieval indexes.

9.2.3. Typed action gateway

Actions are structured requests, not free-form tool invocations. The gateway validates schemas, release identity, job scope, object allowlists, budgets, approvals, evidence prerequisites and expiration. It should issue single-use capability tokens rather than reveal reusable credentials.

A canonical action envelope is

```
ActionRequest {
  actor_release, mission_id, job_id, action_type,
  target_object, parameters, purpose, evidence_refs,
  requested_effect, budget_claim, approval_refs, expires_at
}
```

The gateway returns a signed action result containing executed parameters, external identifiers, state delta, cost, telemetry and rollback handle.

9.2.4. Scenario and adversarial engine

The scenario engine samples from ρ and composes ordinary, edge, tail and adversarial events. It should support:

- ▶ fixed replay seeds and protected random seeds;
- ▶ causal perturbations, not only superficial prompt variations;
- ▶ correlated failures and regime shifts;
- ▶ strategic agents whose policies adapt;
- ▶ tool, provider, sensor and communication outages;
- ▶ authority traps and score-gaming opportunities;
- ▶ transfer across preregistered mission families.

Scenario generation can be model-assisted, but proposed scenarios do not become authoritative until reviewed, versioned and placed under appropriate custody.

9.2.5. Independent verifier service

The verifier service combines heterogeneous methods:

$$\mathcal{V} = \mathcal{V}_{\text{det}} \oplus \mathcal{V}_{\text{stat}} \oplus \mathcal{V}_{\text{model}} \oplus \mathcal{V}_{\text{human}} \oplus \mathcal{V}_{\text{prod}}. \quad (99)$$

It assesses mission outcome, evidence coverage, calibration, critical errors, policy compliance, human rescue, complete cost and production transfer. Each verifier has a scope, calibration record, known failure modes and independence metadata. Disagreements are preserved rather than silently averaged.

9.2.6. Proof-receipt service

Every episode emits a receipt with:

- ▶ environment, constitution and candidate versions;
- ▶ scenario, seed and protected-case identifier;
- ▶ role observations and actions subject to rights policy;
- ▶ tool calls and external effects;
- ▶ human interventions;
- ▶ reward vector and constraint events;
- ▶ verifier outputs and dispute state;
- ▶ complete cost and latency;
- ▶ termination or truncation reason;
- ▶ hashes, signatures and timestamps;
- ▶ evidence-retention and access policy.

Receipts should support deterministic replay where possible and evidentiary reconstruction where exact replay is impossible.

9.2.7. Authority control plane

The authority plane enforces action masks, spend, rates, identities, approvals, network routes, time-to-live, anomaly rules, kill switches and rollback. It consumes signed admission records and emits machine-enforceable credentials. It should remain operational when the candidate is unavailable and should not depend on the candidate to report its own violation.

9.2.8. Chronicle

Chronicle stores accepted capabilities, failures, decisions, conditions, lineages and invalidations. It is append-oriented, versioned and policy-governed. It distinguishes raw logs, working evidence, signed proof and canonical Chronicle entries. Not every generated artifact deserves permanent memory.

9.3. Trust boundaries

Table 13: Minimum trust-boundary rules.

Boundary	Required property
Formation → Proof	Read-only submission of frozen release; no protected-case discovery; no scorer modification.
Proof → Admission	Signed scorecard and evidence references; no credential issuance by verifier.
Admission → Authority	Machine-readable grant tied to exact release, mission, objects, limits and expiry.
Candidate → Tools	Typed, least-privilege, single-use actions through external gateway.
Production → Chronicle	Incident and outcome evidence admitted under rights, validity and challenge rules.
Chronicle → Foundry	Controlled retrieval of valid knowledge and failures; no hidden-proof leakage.
Provider → Candidate	Dependency manifest, version pinning where possible, change detection and fallback.
Human → Candidate	Authenticated role, attributable intervention, explicit approval semantics and non-repudiation.

9.4. Event sourcing and lineage

An event-sourced architecture is natural because proof and rollback depend on ordered state transitions. Let each event be

$$e_k = (\text{id}, t, \text{actor}, \text{release}, \text{mission}, \text{job}, \text{type}, \text{payloadHash}, \text{prevHash}, \text{signature}). \quad (100)$$

The hash-linked event stream supports tamper evidence, but cryptography does not establish semantic truth. The verifier and governance process still determine whether an event supports a claim.

Lineage should connect:

$$\begin{array}{c} \text{Constitution} \rightarrow \text{Environment} \rightarrow \text{Candidate} \rightarrow \text{Run} \rightarrow \text{Proof} \\ \downarrow \\ \text{Chronicle} \leftarrow \text{Outcome} \leftarrow \text{Authority} \leftarrow \text{Admission}. \end{array} \quad (101)$$

This chain makes every operating privilege traceable to current evidence and every successor claim traceable to the exact work on which it was proved.

9.5. Security and resilience profile

A production implementation should include:

- ▶ isolated execution for untrusted candidate code and model tools;
- ▶ secret management outside prompts and candidate memory;
- ▶ egress filtering and destination allowlists;
- ▶ dependency pinning, software bills of materials and provenance;
- ▶ multi-provider or local fallback for mission-critical components;
- ▶ encrypted data at rest and in transit with role-specific keys;
- ▶ independent audit logs and protected evidence storage;
- ▶ tested backup, restore, rollback and disaster-recovery procedures;
- ▶ rate, spend and exposure limits at infrastructure level;
- ▶ incident severity taxonomy, on-call ownership and revocation drills.

The architecture should assume that models, providers, humans and verifiers can fail simultaneously in correlated ways. Resilience must therefore include simple deterministic fallbacks and the retained capacity to return work to the incumbent.

9.6. Environment conformance testing

Before candidate evaluation, the Gym itself should pass a conformance suite:

1. reset and seed behavior is reproducible where specified;
2. observations conform to role schemas and do not leak protected state;
3. invalid actions are rejected externally;
4. termination and truncation are distinguishable;
5. budgets and clocks cannot underflow or be bypassed;
6. proof receipts are complete and hash-consistent;
7. verifier disagreements and failures are preserved;
8. rollback restores a known-good state;
9. environment version changes invalidate or migrate evidence explicitly;
10. protected cases remain inaccessible to formation identities.

Architecture invariant

The same release identity must be visible in the candidate manifest, proof receipt, admission record, Authority Envelope, telemetry stream, incident record and Chronicle lineage. An alias or product name is not enough.

9.7. Minimal viable implementation

The first production-quality implementation need not contain every advanced component. It must contain the separations:

1. one narrow mission and versioned Constitution;
2. one replayable environment with typed observations and actions;
3. incumbent and credible alternatives;
4. one isolated candidate formation workspace;
5. one frozen release and protected case set;
6. independent verification and signed scorecard;

7. one reversible canary with external authority controls;
8. proof receipts, Chronicle entry, rollback and revocation test.

Everything else should be added in response to measured proof and operational debt, not interface ambition.

10. One real succession cycle

The next legitimate threshold is not another conceptual layer. It is one real succession cycle in which a mission is frozen, a Gym is executed, competing complete architectures are formed, one challenger is frozen, protected comparison is completed and the institution either admits a bounded successor or records a clean evidence-backed rejection.

10.1. The empirical completion gate

A real cycle is complete only when all of the following exist:

1. a signed Mission Constitution with exact beneficiary, incumbent, credible alternatives, constraints and authority ceiling;
2. a versioned Mission Gym with replay, edge cases, adversarial cases and conformance tests;
3. a measured incumbent baseline and complete cost model;
4. a documented Successor Manifold and SEIZE underwriting decision;
5. bounded AGI Jobs with typed actions, verifier and proof receipts;
6. at least two materially credible candidate architectures or one candidate plus a strong incumbent and external alternative;
7. one frozen challenger release with a complete dependency manifest;
8. a protected Fresh-Proof Gym and independent custody;
9. a signed comparative scorecard with uncertainty, critical failures, interventions and complete burden;
10. an accountable decision: admit, limited admit, shadow, repair, hold, reject or retire;
11. if admitted, one reversible canary, an exact Authority Envelope, rollback and revocation drill;
12. Chronicle admission of the result, including failures, dissent and the next requalification trigger.

A candidate rejection is a successful cycle when the decision is timely, evidence-backed and preserves learning. The purpose of the Gym is not to manufacture a winner regardless of evidence.

10.2. Recommended first mission

The fastest technical proof is a **Financial Reconciliation or Spreadsheet Integrity Successor Gym** because it offers:

- ▶ repeatable historical cases;
- ▶ deterministic accounting and structural constraints;
- ▶ known resolutions and rapid scoring;
- ▶ low initial external authority;
- ▶ clear comparison against humans, general models and deterministic tools;
- ▶ natural evidence receipts and replay.

A strong physical-enterprise flagship is a **Portfolio Building Operations Successor Gym**: digital twins, sensor streams, repeated failure classes, simulatable interventions, measurable economic value and eventual bounded physical authority. A second-stage strategic product is an **Acquisition Underwriting Gym** focused initially on source-grounded reconstruction, contradiction detection, active information acquisition, thesis versus anti-thesis and uncertainty calibration. It should not pretend that a simulator can perfectly reveal long-term acquisition success.

10.3. The 90-day program

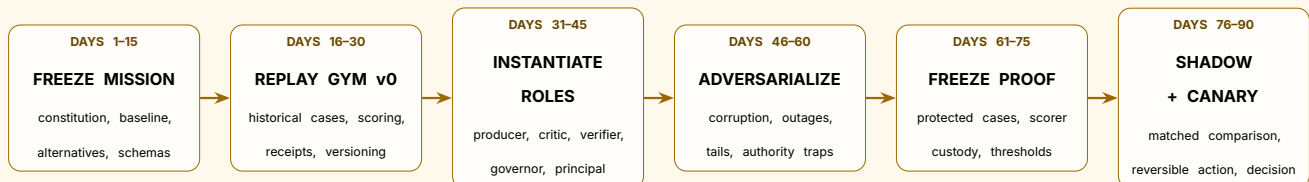


Figure 10: A practical 90-day path to one real succession cycle. The exact duration varies, but the ordering and custody boundaries should remain.

10.3.1. Days 1–15: freeze the mission

Create the Mission Constitution, incumbent baseline, best credible alternative set, state and observation schema, action catalogue, critical failures, proof budget and authority ceiling. The mission should be narrow enough that success and failure are independently knowable.

10.3.2. Days 16–30: build Replay Gym v0

Implement historical episodes, deterministic or adjudicated scoring, baseline policies, proof receipts, environment versioning and conformance tests. Measure the incumbent before optimizing the challenger.

10.3.3. Days 31–45: instantiate institutional roles

Add producer, critic, verifier, governor and human principal, each with role-specific observations and credentials. Record human interventions and decision latency.

10.3.4. Days 46–60: adversarialize

Add corrupted inputs, tool outages, difficult tails, reward-hacking tests, authority-violation traps, fallback scenarios and strategic counterparties. Record which failures reflect the candidate and which expose defects in the Gym.

10.3.5. Days 61–75: freeze Fresh Proof

Separate protected cases, scorer internals, candidate credentials, evaluator custody and promotion thresholds. Freeze one challenger and the comparison set.

10.3.6. Days 76–90: shadow and canary

Run matched incumbent comparison, shadow operation and one reversible bounded action. Produce an admission, repair, hold or stop decision. Test rollback and revocation even when the canary succeeds.

10.4. Acceptance criteria

The cycle should preregister a scorecard with four classes of criteria.

Table 14: Acceptance criteria for one real cycle.

Class	Example criteria	Decision logic
Mission dominance	Positive lower-bound margin over incumbent and best credible alternative; complete cost and rescue included.	Must exceed θ_m .
Critical ceilings	Unauthorized action, rights violation, catastrophic error, uncontained escape, unlogged intervention.	Zero tolerance or explicit constitution ceiling.
Transfer and resilience	Unseen case families, provider outage, degraded data, tail regime, rollback and surrender of control.	Non-inferiority or mission-specific floor.
Institutional readiness	Release identity, telemetry, evidence custody, approval, canary, expiry, rollback and revocation.	Required for authority level; otherwise shadow or repair.

10.5. Illustrative synthetic cycle

Consider a synthetic reconciliation mission with 2,400 replay cases, 400 protected Fresh-Proof cases and 80 adversarial cases. The numbers below demonstrate reporting structure only.

Table 15: Synthetic matched Fresh-Proof scorecard; no empirical claim.

Metric	Incumbent	General model	Hybrid H1	Frozen H2
Verified completion rate	92.0%	93.5%	97.0%	97.8%
Critical-error rate	0.25%	1.00%	0.25%	0.00%
Median cycle time	38 min	9 min	12 min	8 min
Human rescue minutes/case	11.0	7.5	4.0	2.5
Evidence completeness	86%	74%	96%	99%
Outage completion rate	89%	31%	86%	93%
Transfer completion rate	90%	84%	94%	96%
Rollback success	N/A	100% sandbox	100%	100%

Suppose H2's risk-adjusted spread clears the threshold and all hard gates, but production basis risk remains material. The correct decision is *limited admission at A3 for one reversible job class*, not general production authority. After the canary, the institution either expands within the same class, holds, repairs or revokes.

10.6. Evidence package

The final empirical package should contain:

- ▶ Mission Constitution and change log;
- ▶ Gym source, environment manifest and conformance report;
- ▶ scenario catalogue with protected/public designations;
- ▶ incumbent and alternative baselines;
- ▶ candidate release manifest and software bill of materials;
- ▶ preregistration and statistical analysis plan;
- ▶ run receipts, intervention ledger and cost ledger;
- ▶ signed Fresh-Successor Scorecard;
- ▶ admission record and dissent;
- ▶ Authority Envelope, canary plan and rollback proof;
- ▶ Chronicle entries, impairment record, requalification date and revocation drill;
- ▶ package-wide hashes and reproducibility instructions.

10.7. What failure looks like

The cycle is incomplete when:

- ▶ the mission remains broad or aspirational;
- ▶ the incumbent was never measured;
- ▶ the same cases were used for formation and promotion;
- ▶ the candidate changed after protected cases were observed;
- ▶ costs or human rescue are omitted;
- ▶ the verifier is not independent;
- ▶ the score is positive but the hard gates were never defined;
- ▶ production authority exists without exact identity, rollback or expiry;
- ▶ the decision is not preserved in Chronicle;
- ▶ a failed candidate is quietly replaced without recording the failure.

Empirical completion

A real succession cycle ends with one of two legitimate outcomes:

**a Mission-Dominant Specialist ASI admitted under bounded authority,
or a clean evidence-backed rejection that improves the next cycle.**

11. Sovereign assets, economics and maturity

The enterprise may rent models while owning the mission. The strategically durable asset is not one model release. It is the executable constitution, rights-cleared environment, protected comparison,

proof receipts, Authority Envelopes and Chronicle that make models substitutable competitors inside an institution-owned proving system.

11.1. The Gym as a sovereign corporate asset

The institution should seek ownership or durable control over:

- ▶ the Mission Constitution and beneficiary definition;
- ▶ environment definitions and transition models;
- ▶ rights-cleared scenario data and case families;
- ▶ typed action schemas and workflow integration;
- ▶ protected cases and evaluator infrastructure;
- ▶ failure taxonomies and adversarial scenarios;
- ▶ proof receipts, scorecards and intervention ledgers;
- ▶ Chronicle, lineage and requalification history;
- ▶ Authority Envelopes, credentials and rollback state;
- ▶ portability interfaces that allow model replacement.

A model can then be replayed through the same Gym. The institution can ask whether a new model improves the current successor, introduces a new failure, permits a smaller local architecture, reduces provider Beta or leaves the incumbent superior. Provider bargaining power changes because the enterprise owns the mission-specific basis of comparison.

11.2. Complete economics

Let the total economic value of candidate x be

$$\text{TEV}_m(x) = V_{\text{direct}} + V_{\text{risk}} + V_{\text{option}} + V_{\text{learning}} + V_{\text{sovereignty}} - C_{\text{build}} - C_{\text{run}} - C_{\text{switch}} - C_{\text{control}} - C_{\text{failure}}. \quad (102)$$

Direct value includes mission outcome, throughput and quality. Risk value includes avoided loss and faster detection. Option value includes portability, local fallback and future model substitution. Learning value includes reusable cases, verifier calibration and Chronicle. Sovereignty value includes reduced lock-in and retained authority. Costs include formation, integration, operation, switching, proof, monitoring, rollback, rights and expected failure.

The Gym's value can be approximated as the present value of future comparison and renewal options:

$$V(\mathcal{G}_m) = \sum_{t=0}^T \frac{\mathbb{E}[\Delta \mathcal{S}_{m,t} + Q_{\text{substitution},t} + Q_{\text{risk},t}]}{(1+r)^t} - C_{\mathcal{G},t}. \quad (103)$$

A high-quality Gym may remain valuable even when the first challenger fails, because it makes the incumbent measurable, exposes failure modes, creates reusable proof capital and improves the next architecture search.

11.3. Requalification as a recurring institution

Every material change to model, data, tool, workflow, verifier, provider, mission distribution or authority creates a legitimate requalification event. This supports a recurring operating model based on evidence freshness rather than artificial subscription mechanics. The recurring deliverables are:

- ▶ environment updates and new adversarial scenarios;
- ▶ incumbent and challenger replays;
- ▶ provider and dependency delta tests;
- ▶ authority renewal or contraction;
- ▶ Chronicle maintenance and invalidation;
- ▶ fresh canaries and rollback drills;
- ▶ proof-debt retirement.

11.4. Customer-facing product and institutional stack

The customer-facing object should remain the **GoalOS Mission-Sovereign Successor**. The customer does not primarily want to buy a benchmark. The Gym is the factory, proving range and renewal infrastructure behind the operating institution.

A coherent stack is:

1. **GoalOS Mission Gym Builder**: converts the Mission Constitution into an executable environment.
2. **GoalOS Successor Foundry**: forms and searches competing complete architectures.
3. **GoalOS Fresh-Proof Arena**: protects independent comparative evaluation.
4. **GoalOS Authority Simulator**: rehearses approval, refusal, escalation, failure and rollback.
5. **GoalOS Successor Operations**: runs the admitted system under bounded authority.
6. **GoalOS Chronicle and Requalification**: preserves outcomes and continually challenges the incumbent successor.

The stack is valuable because it connects advisory reasoning to executable work, protected evidence, operational authority and renewal. Each component should remain independently inspectable and replaceable.

11.5. Maturity model

Table 16: GoalOS maturity model.

Level	State	What exists
L0 — Advisory interface	Structured reasoning	Mission framing, thesis, anti-thesis and recommendation; no executable environment.
L1 — Replay laboratory	Historical episodes	Fixed cases, baselines, scoring and proof receipts.
L2 — Interactive Mission Gym	State transitions	Candidate chooses information and actions over time.
L3 — Multi-agent Successor Gym	Institutional roles	Producer, critic, verifier, governor and human principal.

Level	State	What exists
L4 — Protected Fresh-Proof Plane	Hidden evaluation	Frozen challenger, blind cases, critical gates and independent custody.
L5 — Canary Successor	Bounded reality	Shadow and reversible live action with external control.
L6 — Mission-Sovereign Successor	Admitted institution	Identity, scoped authority, Chronicle, monitoring, requalification and revocation.
L7 — Recursive Succession Institution	Successor to the successor	Continual architecture competition, protected proof and governed renewal.

A rich prompt interface is not yet a Gym. A benchmark is not yet a Successor. A high-scoring Successor is not yet Mission-Sovereign. Maturity should be claimed only when the corresponding artifacts and controls exist.

11.6. Capital allocation across missions

Let mission m have expected option value O_m , Gymmability G_m , strategic importance S_m , time-to-proof T_m , proof cost C_m and containment risk R_m . A portfolio priority score can be

$$P_m = \frac{O_m^{w_o} G_m^{w_g} S_m^{w_s}}{T_m^{w_t} C_m^{w_c} (1 + R_m)^{w_r}}, \quad (104)$$

subject to minimum rights, data and governance readiness. The score is not a substitute for judgment. It disciplines the common error of beginning with the most visible mission rather than the mission most likely to produce trustworthy proof.

A portfolio should contain:

- ▶ one fast high-Gymmability proof mission;
- ▶ one strategic mission with larger economic upside;
- ▶ one physical or consequential mission developed first at low authority;
- ▶ one reserve mission that becomes feasible as instrumentation improves.

11.7. Build, rent, partner or acquire

The Successor Manifold includes transaction choices. A build decision is favored when mission knowledge, data, workflow and proof infrastructure are differentiating and reusable. Rent is favored when provider capabilities are commoditized and portability is strong. Partnership is favored when rights, integration or domain expertise are complementary. Acquisition is favored when the target owns scarce data, workflow position, talent or customer access that cannot be replicated economically. The Gym provides a common proving environment for all four choices.

11.8. Economic anti-patterns

False economy

The following practices systematically overstate value:

- ▶ counting model-token cost while omitting integration, review, incident and switching cost;
- ▶ treating hidden human rescue as free;
- ▶ comparing the candidate to a weak or outdated incumbent;
- ▶ ignoring the option value of local fallback and provider substitution;
- ▶ granting broad authority to avoid the engineering cost of typed actions;
- ▶ postponing proof until after deployment;
- ▶ preserving only successes in institutional memory.

11.9. The strategic result

Sovereignty doctrine

An institution may rent general intelligence while owning the mission constitution, bounded jobs, proprietary complements, evidence, Chronicle, permissions, portability and economic capture. The Gym makes external models compete inside an institution-owned reality; Successor Ω turns the winning architecture into bounded and renewable operating capacity.

12. Research program and final judgment

This paper proposes an executable institutional architecture. Its central claims are structural: the Gym, candidate and admitted successor are distinct; comparative proof and authority must be separated; value must be assessed against credible alternatives after complete burden and basis risk; and recursive improvement must not become self-authorization. These claims can be falsified, refined and operationalized only through real succession cycles.

12.1. Research agenda

12.1.1. Mission-environment science

1. Develop estimators and confidence bounds for Gym basis risk before extensive production data exist.
2. Identify minimal sufficient state representations that preserve transfer while respecting privacy and rights.
3. Measure Gymmability and determine when a mission should remain advisory rather than executable.
4. Construct scenario distributions that represent ordinary work, tails, adversarial adaptation and endogenous market response.
5. Develop environment conformance tests for institutional missions, not only software APIs.

12.1.2. Proof science

1. Quantify evaluator correlation, effective verifier count and the value of methodological diversity.
2. Design sequential Fresh-Proof procedures that preserve validity under repeated monitoring and candidate turnover.
3. Separate superiority, non-inferiority and hard-gate evidence in one interpretable scorecard.
4. Estimate contamination from public benchmarks, provider training and repeated exposure.
5. Develop proof-debt measures that predict when apparent confidence exceeds evidentiary support.

12.1.3. Institutional game theory

1. Model incumbent resistance, challenger hype, provider lock-in and evaluator effort in one repeated Bayesian game.
2. Design mechanisms that reward truthful uncertainty, justified abstention and complete intervention reporting.
3. Test collusion and correlated-failure defenses across producer, critic, verifier and governor roles.
4. Evaluate strategic counterparties whose behavior changes in response to the successor's presence.
5. Determine which protected-proof rules remain credible when failure modes evolve faster than fixed test suites.

12.1.4. Authority and control

1. Formalize viability kernels and barrier functions for money, rights, safety, privacy and governance.
2. Calibrate authority expansion and contraction as a control process coupled to evidence and monitoring capacity.
3. Define minimal reversible canaries for missions with delayed or partially irreversible effects.
4. Test credential invalidation, route closure, rollback and fallback under correlated outage.
5. Measure whether candidates learn to abstain, escalate and surrender control under realistic pressure.

12.1.5. Recursive succession

1. Identify constitutional invariants that must survive every architecture generation.
2. Study how Chronicle inheritance affects innovation, contamination, lock-in and proof leakage.
3. Develop institution-level scaling laws for mission value, proof cost, control burden and successor turnover.
4. Detect recursive degradation when each local replacement appears beneficial.
5. Characterize stable renewal regimes in which architecture changes rapidly while accountability remains continuous.

12.2. Limitations

Several limits should remain explicit.

First, no formal environment fully captures an open world. Gym basis risk can be reduced, bounded and monitored, not abolished. Second, independent verification itself can fail, collude or share model and data blind spots. Third, a numerical Sovereignty Spread depends on constitutional choices about value, rights and burden; it does not eliminate judgment. Fourth, some missions have slow, confounded or disputed outcomes and should not be forced into false precision. Fifth, high performance in one mission does not imply general superiority or transferable authority. Sixth, the paper does not claim

that a 90-day program will always produce a viable successor; it claims that the program can produce a legitimate decision and a reusable evidence institution.

12.3. The empirical completion gate

The decisive next step is one real succession cycle. The minimum publishable evidence is:

one frozen mission, one measured incumbent, one executable Gym, one frozen challenger, one independent fresh-work result, and one accountable admit, repair, hold or reject decision.	(105)
---	-------

If authority is granted, the cycle must also contain one reversible canary, exact release identity, an Authority Envelope, telemetry, rollback, expiry, revocation and Chronicle admission.

12.4. Final synthesis

The Gym supplies:

- ▶ the mission's operational world;
- ▶ repeatable experience and counterfactuals;
- ▶ adversity and strategic response;
- ▶ comparative measurement;
- ▶ authority rehearsal;
- ▶ fresh requalification.

The Specialist ASI supplies:

- ▶ the complete candidate intelligence architecture;
- ▶ models, data, tools, workflow and humans;
- ▶ evidence production and verifier interaction;
- ▶ resilience, integration and economic performance;
- ▶ a frozen object capable of comparative proof.

Successor Ω supplies:

- ▶ accountable identity and admission;
- ▶ scoped, monitored and revocable authority;
- ▶ Chronicle-backed memory and continuity;
- ▶ rollback, expiry and requalification;
- ▶ the governed capacity to prepare the next successor.

The complete institution is

GoalOS Successor System = Mission Gym + Successor Foundry + Independent Fresh Proof + Authority Envelope + Chronicle + Accountable Renewal.	(106)
--	-------

The decisive distinction is

Specialist ASI = the complete candidate that proves it can win,
 Mission-Sovereign Successor = the proven winner the institution admits to act,
 Successor Ω = the admitted winner bound to governed renewal.

(107)

Definitive formulation

The Mission Gym makes the mission executable.
The Successor Foundry manufactures competing institutions.
Fresh Proof determines which one actually wins.
The Authority Envelope determines how the winner may enter reality.
Chronicle determines what survives long enough to compound.

Or, most compactly:

The Gym teaches and tests. Reality proves.
 The institution admits. Chronicle prepares the successor.

(108)



Test. Prove. Admit. Govern. Renew.

A. Formal definitions and propositions

A.1. Mission Constitution

Mission Constitution

A Mission Constitution is

$$\mathcal{C}_m = (m, b, h, \mathcal{B}, \mathcal{I}, \mathcal{W}, \mathcal{P}, \mathbf{q}, \mathbf{c}, K, A_{\max}, \mathcal{E}, \mathcal{X}), \quad (109)$$

where m is the exact mission, b the beneficiary, h the horizon, $\mathcal{B}$ the baseline and credible alternatives, $\mathcal{I}$ permitted information, $\mathcal{W}$ permitted work, $\mathcal{P}$ prohibited actions, $\mathbf{q}$ quality criteria, $\mathbf{c}$ critical constraints, K proof capital, $A_{\max}$ authority ceiling, $\mathcal{E}$ evidence rules and $\mathcal{X}$ expiry and requalification conditions.

A.2. Admissible candidate

A candidate x is admissible if

$$\text{Admissible}_{\mathcal{C}_m}(x) = 1 \quad (110)$$

only when its data, tools, providers, human roles, rights, security, fallback and intended authority satisfy pre-proof feasibility. Admissibility is distinct from performance.

A.3. Best credible alternative

Let $\mathcal{B}_m$ contain alternatives that are operationally available or reasonably procurable within the decision horizon. Alternative y is credible when its capability, cost, rights, integration and delivery assumptions are evidence-backed. The best credible alternative is scenario-dependent:

$$B_m(\omega) = \arg \max_{y \in \mathcal{B}_m} V_m(y; \omega). \quad (111)$$

A weak straw baseline invalidates Mission Alpha.

A.4. Fresh mission dominance

Mission-dominant Specialist ASI

A frozen candidate $x^\star$ is a Mission-Dominant Specialist ASI at confidence level $1 - \delta$ when:

1. $x^\star$ is admissible and exactly identified;
2. protected comparison covers the preregistered mission family;
3. $\mathcal{S}_m(x^\star) > \theta_m$ under a valid lower-bound procedure;
4. $\mathbf{c}(x^\star) = \mathbf{0}$ and all critical ceilings are met;
5. no omitted credible alternative is reasonably likely to reverse the decision;
6. the scorecard includes complete burden, intervention and basis-risk treatment.

A.5. Accountable admission

Admission

Admission is an externally authorized, signed and time-bounded institutional act that maps a proven release and current evidence into an operating state and exact Authority Envelope:

$$\text{Adm} : (x^\star, \mathcal{C}_m, E_t, H_t) \mapsto (d, A, T, \mathcal{Q}), \quad (112)$$

where d is the decision, A authority, T expiry and $\mathcal{Q}$ monitoring, requalification and revocation conditions.

A.6. Mission-Sovereign Successor and Successor Ω

Mission-Sovereign Successor

A Mission-Sovereign Successor is a Mission-Dominant Specialist ASI after admission, with exact release identity, a scoped Authority Envelope, monitoring, rollback, expiry, revocation and Chronicle linkage.

Successor Ω

A Mission-Sovereign Successor becomes Successor Ω when the operating institution also contains a governed, externally proved and non-self-authorizing process for generating, comparing, admitting and handing over to its successor.

A.7. Authority non-monotonicity

Proposition A.1 (Evidence-conditioned authority). *Authority is monotone in neither capability nor elapsed time. For fixed mission and risk class, authority may expand only after sufficient evidence and control capacity, and must contract when evidence freshness, monitoring, rollback or environmental fit deteriorates.*

A generic authority update is

$$A_{t+1} = \text{Clamp}_{[0, A_{\max}]} (A_t + \gamma_+ \Delta E_t + \gamma_K \Delta K_t - \gamma_D \Delta D_t - \gamma_I I_t), \quad (113)$$

subject to immediate hard-stop revocation. Here E is evidence, K control capacity, D proof debt and I incident severity.

A.8. Proof freshness

Let evidence item e_j have validity function $w_j(t) \in [0, 1]$. A proof state is current when

$$\sum_j w_j(t) q_j \geq Q_{\min} \quad (114)$$

and no required item has expired. Some evidence is conjunctive rather than additive: absence of a current rights attestation, release identity or rollback proof can invalidate authority regardless of aggregate score.

A.9. Basis-risk-adjusted lower bound

Suppose the protected estimate of candidate advantage is $\hat{\Delta}$ with one-sided uncertainty radius u_δ , and estimated Gym basis reserve is $\hat{\epsilon}$ with reserve b_δ . Then

$$\underline{\Delta}_{1-2\delta}^{\text{real}} = \hat{\Delta} - u_\delta - \hat{\epsilon} - b_\delta \quad (115)$$

is a conservative real-world lower bound under the stated assumptions. Promotion requires

$$\underline{\Delta}_{1-2\delta}^{\text{real}} > \theta_m. \quad (116)$$

The form is intentionally transparent: statistical uncertainty and environment discrepancy are charged separately.

A.10. No-free-succession proposition

Proposition A.2 (No free succession). *If candidate replacement has switching cost $C_s > 0$, integration risk $R_i > 0$ or authority requalification cost $C_a > 0$, then a positive raw performance difference is insufficient for rational succession. The robust comparative margin must exceed all justified replacement burdens.*

A.11. Non-compensation proposition

Proposition A.3 (Lexical constraints). *If a constraint c_j is constitutionally non-compensable, no finite increase in scalarized mission value can make a candidate satisfying $c_j > 0$ eligible for admission.*

This prevents a large revenue or speed benefit from purchasing permission to violate rights, safety or authority.

A.12. Recursive invariance

Let $\mathcal{I}(\Omega_t)$ be the set of constitutional invariants present at epoch t . A recursive system is valid when

$$\mathcal{I}_{\min} \subseteq \bigcap_{t \geq 0} \mathcal{I}(\Omega_t), \quad (117)$$

with

$$\mathcal{I}_{\min} = \left\{ \begin{array}{l} \text{independent proof, exact identity, bounded authority,} \\ \text{rollback, revocation, Chronicle integrity, accountable admission} \end{array} \right\}. \quad (118)$$

A.13. Decision-state machine

The admissible transitions are

$$\text{Proposed} \rightarrow \text{Forming} \rightarrow \text{Frozen} \rightarrow \text{Proved} \rightarrow \text{Admitted} \rightarrow \text{Operating} \rightarrow \text{Requalified} \quad (119)$$

with side transitions to Repair, Hold, Reject, Rollback, Revoke and Retire. Forbidden transitions include Proposed $\rightarrow$ Operating, Forming $\rightarrow$ Admitted and candidate-controlled Proved $\rightarrow$ Admitted.

B. Canonical algorithms

Input: Environment release $\mathcal{G}_m^v$, candidate release x^r , scenario policy ρ , authority level A_k

Output: Proof receipt R_e

Verify hashes, rights, credentials and compatibility;

Sample scenario and initialize state s_0 under protected seed;

Initialize clocks, budgets, telemetry and intervention ledger;

for $t = 0, 1, \dots$ *until ending condition* **do**

 Generate role-specific observations $o_t^i = O_i(s_t, h_t)$;

 Obtain candidate and institutional actions $\mathbf{a}_t$;

 Validate each action against schema, job, budget and Authority Envelope;

if *action is admissible* **then**

 | execute through typed gateway and record result

else

 | reject, record violation and apply stop/escalation rule

end

 Apply exogenous event and transition $s_{t+1} \sim P(\cdot \mid s_t, \mathbf{a}_t, \xi_t)$;

 Compute value vector $\mathbf{r}_t$ and hard constraints $\mathbf{c}_t$;

 Record human intervention, cost, latency, telemetry and state delta;

if *critical constraint fires* **then**

 | terminate, revoke credentials if required, preserve evidence;

end

if *mission success or failure predicate holds* **then**

 | terminate with typed reason;

end

if *time, budget or authority window expires* **then**

 | truncate with typed reason;

end

end

Run independent verifier ensemble and dispute procedure;

Construct, sign and store R_e with hashes, endings and custody metadata;

return R_e ;

Algorithm 1: Mission Gym episode

C. Canonical evidence and control schemas

The following schemas are normative design templates, not language-specific APIs. Fields may be implemented in JSON, protocol buffers, relational tables or event streams, but identity, versioning, provenance and status semantics should remain explicit.

Input: Mission Constitution $\mathcal{C}_m$, candidate manifold $\mathfrak{X}_m$, evidence state E , proof budget K

Output: Frozen challenger $\hat{x}$ or stop decision

Measure incumbent and validate best credible alternative set;

Remove candidates failing pre-proof admissibility;

Estimate Mission Alpha, Beta, complete burden, tail risk and basis-risk reserve;

Compute provisional Sovereignty Spread distribution for each candidate;

Construct thesis, anti-thesis and impairment case;

while $K > 0$ *and no candidate is freeze-ready* **do**

 Enumerate feasible experiments $e \in \mathcal{E}$;

 Estimate $\text{EVSI}(e)$, cost, risk and time;

 Select e^* by Equation (68);

 Execute experiment in Formation/Adversarial/Transfer Gym;

 Update evidence, candidate architectures and proof debt;

 Apply stop, repair and reserve rules;

end

if *one candidate has positive freeze-ready spread and no unresolved blocker* **then**

 freeze exact release, dependencies, metrics and protected comparison plan

else

 return hold, repair or reject with evidence

end

return $\hat{x}$ or decision;

Algorithm 2: SEIZE proof-capital allocation

Input: Frozen challenger $\hat{x}$, incumbent I , alternatives $\mathcal{B}$, protected Gym $\mathcal{G}_m^{\text{fresh}}$, preregistration Π

Output: Signed Fresh-Successor Scorecard S

Verify candidate and environment hashes; confirm no protected-case access;

Run matched episodes for $\hat{x}$, I and $\mathcal{B}$ under blinded custody;

Collect receipts, interventions, costs, endings and verifier outputs;

Estimate comparative Alpha and uncertainty under Π ;

Estimate Beta, tail loss, basis-risk reserve and proof debt;

Evaluate hard constraints, transfer, resilience, rights and governance;

Compute Sovereignty Spread and sensitivity analysis;

Record disagreements, excluded cases, contamination and protocol deviations;

Sign scorecard with evaluator identities and evidence references;

return S ;

Algorithm 3: Fresh-Successor Gate

Input: Scorecard S , Mission Constitution $\mathcal{C}_m$, proposed authority A , production readiness P , accountable principal H

Output: Admission record D and optional Authority Envelope $A^\star$

Confirm H is authorized and conflicts are disclosed;

Verify scorecard signatures, freshness, release identity and critical gates;

Review mission dominance, uncertainty, basis risk, dissent and credible alternatives;

if *hard gate failed or evidence invalid* **then**

 set decision Reject or Repair; issue no credentials;

else if *dominance not established* **then**

 set decision Hold, Shadow or Repair;

else

 choose least authority sufficient for proved mission value;

 define objects, actions, budgets, approvals, telemetry, expiry and fallback;

 require rollback and revocation proof appropriate to authority class;

 sign admission record and machine-readable Authority Envelope;

 issue short-lived least-privilege credentials through independent control plane;

end

Store decision, dissent and next requalification trigger in Chronicle;

return $D, A^\star$;

Algorithm 4: Accountable admission

Input: Operating successor Ω_t , Authority Envelope A_t , telemetry stream Y_t

Output: Continue, contract, rollback, revoke or requalify

foreach *job or action request* **do**

 verify release identity, mission, object, budget, approval and evidence freshness;

 deny request if any predicate is false;

 execute admissible request through gateway and emit signed result;

 update monitoring, drift, incident and basis-risk state;

if *critical event or credential anomaly* **then**

 revoke, isolate, preserve evidence and return to fallback;

else if *drift, evidence decay or control degradation exceeds threshold* **then**

 contract authority and initiate requalification;

else if *performance and control remain within envelope* **then**

 continue;

end

end

Periodically run canary/requalification Gym and update Chronicle;

Algorithm 5: Continuous operation and fail-closed control

Input: Current successor Ω_t , Chronicle C_t , mission state and opportunity set

Output: Next successor Ω_{t+1} or retained Ω_t

Generate challenger architectures in governed Foundry;

Apply SEIZE underwriting and proof-capital algorithm;

Freeze exactly one challenger before protected cases;

Run Fresh-Successor Gate;

if *challenger clears all gates and accountable admission* **then**

 execute reversible handover and create Ω_{t+1}

else

 retain, repair or retire candidates; keep Ω_t within current envelope

end

Preserve lineage, failures, dissent and decision in Chronicle;

Verify that constitutional invariants survive the transition;

Algorithm 6: Successor-to-Successor cycle

C.1. Environment specification

Listing 1: EnvironmentSpec v1.0

```
EnvironmentSpec {
  environment_id: content_hash
  mission_id: string
  mission_constitution_version: semver
  environment_version: semver
  purpose: formation | adversarial | transfer | fresh_proof | canary | requalification
  state_schema_ref: uri_hash
  role_observation_schemas: map<Role, uri_hash>
  role_action_schemas: map<Role, uri_hash>
  transition_model_ref: uri_hash
  scenario_distribution_ref: uri_hash
  verifier_spec_ref: uri_hash
  reward_vector_schema: uri_hash
  hard_constraint_schema: uri_hash
  termination_conditions: list<TypedCondition>
  truncation_conditions: list<TypedCondition>
  rights_policy_ref: uri_hash
  protected_case_policy: ProtectedCasePolicy
  seed_policy: deterministic | protected_random | externally_sampled
  conformance_report_ref: uri_hash
  created_at: timestamp
  signed_by: list<Signature>
}
```

C.2. Candidate manifest

Listing 2: CandidateManifest v1.0

```
CandidateManifest {
  candidate_release_id: content_hash
  candidate_name: string
  version: semver
  mission_id: string
  model_components: list<ComponentRef>
  data_and_knowledge: list<RightsClearedRef>
  harness_and_orchestration: list<ComponentRef>
  tools_and_adapters: list<ComponentRef>
  verifier_components_used_in_operation: list<ComponentRef>
  workflow_and_human_roles: list<RoleSpec>
  security_resilience_rollback: ControlProfile
  integration_refs: list<ComponentRef>
  initial_chronicle_state_ref: uri_hash
  proposed_authority_class: A0 | A1 | A2 | A3 | A4
  provider_dependencies: list<Dependency>
  software_bill_of_materials_ref: uri_hash
  formation_case_exposure: list<ExposureDeclaration>
  known_limitations: list<ClaimWithEvidence>
  created_at: timestamp
  signed_by: list<Signature>
}
```

C.3. Run proof receipt

Listing 3: RunProof v1.0

```

RunProof {
  run_id: uuid
  mission_id: string
  job_id: string
  environment_id: content_hash
  candidate_release_id: content_hash
  comparison_arm: incumbent | alternative | challenger
  scenario_id: protected_pseudonym
  seed_commitment: hash
  started_at: timestamp
  ended_at: timestamp
  ending_type: success | mission_failure | critical_failure | termination | truncation | operator_stop
  ending_reason_code: string
  observation_log_ref: uri_hash
  action_log_ref: uri_hash
  external_effects: list<SignedActionResult>
  human_interventions: list<Intervention>
  reward_vector: map<Metric, number>
  constraint_events: list<ConstraintEvent>
  verifier_results: list<VerifierResult>
  disputes: list<DisputeRecord>
  complete_cost: CostLedger
  latency_profile: LatencyLedger
  rollback_handle: optional<RollbackRef>
  contamination_status: clean | suspected | exposed | public
  previous_receipt_hash: optional<hash>
  receipt_hash: hash
  signed_by: list<Signature>
}

```

C.4. Fresh-Successor Scorecard**Listing 4:** FreshSuccessorScorecard v1.0

```

FreshSuccessorScorecard {
  scorecard_id: content_hash
  mission_id: string
  constitution_version: semver
  protected_environment_id: content_hash
  candidate_release_id: content_hash
  incumbent_release_id: content_hash
  credible_alternative_release_ids: list<content_hash>
  preregistration_ref: uri_hash
  sampling_and_blocking: AnalysisDesign
  primary_estimands: list<EstimateWithInterval>
  mission_alpha: EstimateWithInterval
  mission_beta: FactorExposureReport
  complete_burden: CostLedger
  tail_risk: RiskReport
  gym_basis_reserve: EstimateWithInterval
  proof_debt: ProofDebtReport
  sovereignty_spread: EstimateWithInterval
  hard_gate_results: list<GateResult>
  transfer_results: list<MetricResult>
  resilience_results: list<MetricResult>
  intervention_report: InterventionLedger
  contamination_and_deviations: list<ProtocolDeviation>
  verifier_independence_report: IndependenceReport
}

```

```

dissent_and_disputes: list<DissentRecord>
conclusion: dominance_established | not_established | invalidated
signed_at: timestamp
signed_by: list<Signature>
}

```

C.5. Admission record and Authority Envelope

Listing 5: AdmissionRecord and AuthorityEnvelope v1.0

```

AdmissionRecord {
  admission_id: content_hash
  scorecard_id: content_hash
  candidate_release_id: content_hash
  mission_id: string
  decision: admit | admit_limited | shadow | repair | hold | reject | retire
  rationale_ref: uri_hash
  dissent_refs: list<uri_hash>
  authority_envelope_id: optional<content_hash>
  effective_at: timestamp
  expires_at: timestamp
  requalification_triggers: list<TypedCondition>
  accountable_principal: Identity
  signed_by: list<Signature>
}

AuthorityEnvelope {
  authority_envelope_id: content_hash
  candidate_release_id: content_hash
  mission_id: string
  authority_level: A0 | A1 | A2 | A3 | A4
  permitted_observations: list<ResourceScope>
  permitted_actions: list<ActionScope>
  permitted_targets: list<ResourceScope>
  budgets: list<BudgetLimit>
  rate_and_exposure_limits: list<Limit>
  required_approvals: list<ApprovalRule>
  jurisdiction_and_rights: list<Condition>
  telemetry_requirements: list<TelemetryRule>
  anomaly_and_fail_closed_rules: list<TypedCondition>
  rollback_and_fallback: RollbackProfile
  credential_policy: CredentialPolicy
  effective_at: timestamp
  expires_at: timestamp
  revocation_authorities: list<Identity>
  signed_by: list<Signature>
}

```

C.6. Chronicle, requalification and revocation

Listing 6: Chronicle and lifecycle records v1.0

```

ChronicleEntry {
  entry_id: content_hash
  mission_id: string
  entry_type: capability | failure | decision | incident | assumption | invalidation | lineage
  subject_release_id: content_hash
  claim: string
}

```

```

evidence_refs: list<uri_hash>
rights_status: RightsStatus
scope: Scope
validity_start: timestamp
validity_end: optional<timestamp>
challenge_status: unchallenged | challenged | resolved | invalidated
lineage_refs: list<content_hash>
admitted_by: list<Signature>
}

RequalificationRecord {
  record_id: content_hash
  mission_id: string
  candidate_release_id: content_hash
  trigger: scheduled | model_change | data_change | tool_change | verifier_change |
    provider_change | mission_shift | authority_change | incident | evidence_expiry
  scope: delta | full
  environment_ids: list<content_hash>
  result: continue | contract | repair | rollback | revoke | retire
  evidence_refs: list<uri_hash>
  new_authority_envelope_id: optional<content_hash>
  signed_by: list<Signature>
}

RevocationRecord {
  record_id: content_hash
  mission_id: string
  candidate_release_id: content_hash
  authority_envelope_id: content_hash
  trigger: string
  severity: critical | high | medium | administrative
  credentials_invalidated_at: timestamp
  routes_closed_at: timestamp
  rollback_result: success | partial | failed | not_applicable
  fallback_owner: Identity
  evidence_preservation_ref: uri_hash
  next_decision: investigate | repair | requalify | retire
  signed_by: list<Signature>
}

```

C.7. Evidence status semantics

Every material claim should have one of the following statuses:

Status	Meaning
Proposed	Generated or asserted; not yet independently supported.
Observed	Directly recorded, but interpretation may remain open.
Verified	Independently checked under a named method and scope.
Contested	A material challenge is unresolved.
Invalidated	Evidence or later observation defeats the claim.
Expired	Previously supported, but validity conditions or time horizon ended.
Canonical	Admitted to Chronicle for a specified scope and validity period.

A user interface should never render Proposed and Verified evidence as visually equivalent.

D. Implementation controls and release checklist

D.1. Minimum accountable roles

Table 17: Minimum role separation for one real cycle.

Role	Core responsibility	Must not control alone
Mission owner	Defines mission, beneficiary, value and operating constraints.	Protected proof, verifier result or re-release credentials.
Formation lead	Builds candidate and Formation Gym.	Protected cases, final scorecard or admission.
Environment custodian	Versions environment and scenario distribution.	Candidate optimization and admission.
Proof custodian	Holds protected cases, runs matched comparison and preserves receipts.	Candidate formation or authority issuance.
Verifier lead	Applies independent verification and documents uncertainty.	Admission credentials or Chronicle alteration.
Governor / accountable principal	Makes admit, hold, repair, reject or revoke decision.	Unlogged changes to proof or candidate release.
Authority operator	Translates admission into credentials, limits, telemetry and kill switches.	Mission dominance determination.
Chronicle custodian	Admits valid memory, failure, lineage and invalidations.	Erasure of contested or adverse evidence.
Incumbent owner	Maintains baseline, fallback and hand-over readiness.	Case selection or obstruction of fair comparison.

Small organizations may combine roles, but conflicts must be disclosed and counter-controls added. The same person can hold several titles; the same credential should not control formation, protected proof and consequential release.

D.2. Mission-freeze checklist

1. Exact mission, beneficiary and horizon are written in one paragraph.
2. Incumbent and best credible alternatives are named and measurable.
3. Success, mission failure, critical failure and administrative truncation are distinct.
4. Permitted information, work, tools and external effects are explicit.
5. Rights, privacy, security and jurisdiction conditions are documented.
6. Value vector and non-compensable constraints are frozen.
7. Proof capital and stop conditions are approved.
8. Maximum authority class for the cycle is set before results are known.
9. Evidence ownership, retention and access are defined.
10. Accountable admission and revocation authorities are named.

D.3. Gym release checklist

1. Environment identity and version are content-addressed.
2. State, observation and action schemas pass conformance tests.
3. Invalid actions are rejected by external infrastructure.
4. Termination and truncation return distinct reason codes.
5. Budgets, clocks and exposure limits cannot be bypassed.
6. Formation, adversarial, transfer and Fresh-Proof case classes are labeled.
7. Protected seeds and scorer internals are inaccessible to formation identities.
8. Every run emits a complete proof receipt.
9. Replay and nondeterminism policies are documented.
10. Known simulator omissions and basis-risk hypotheses are recorded.

D.4. Candidate-freeze checklist

1. Exact model, data, harness, tools, workflow and verifier components are listed.
2. Provider-side mutable dependencies are identified.
3. Rights-cleared data and retrieval indexes are pinned or versioned.
4. Human roles and intervention channels are declared.
5. Known limitations and formation-case exposure are signed.
6. Software bill of materials and security scan are attached.
7. Rollback and fallback paths exist for proposed authority class.
8. Candidate release hash is generated before protected proof begins.
9. No post-freeze change is allowed without creating a new release.
10. Formation credentials cannot access protected cases or authority issuance.

D.5. Fresh-Proof checklist

1. Preregistration defines primary estimands, gates, sample, stopping and exclusions.
2. Incumbent and credible alternatives are versioned.
3. Cases are representative and paired or blocked where feasible.
4. Human interventions, costs, latency, failures and abstentions are logged.
5. Contamination status is recorded for every case.
6. Verifier independence and shared dependencies are assessed.
7. Statistical uncertainty and multiple-selection effects are treated.
8. Gym basis-risk reserve and unresolved proof debt are explicit.
9. Scorecard includes dissent and protocol deviations.
10. Scorecard is signed before admission discussion begins.

D.6. Admission and canary checklist

1. Accountable principal verifies authority and conflicts.
2. Least authority sufficient for the proved value is chosen.
3. Authority Envelope specifies actions, objects, budgets, approvals, expiry and fallback.
4. Credentials are short-lived, least-privilege and independently issued.
5. Telemetry and anomaly rules are active before the first action.
6. Canary is reversible and limited to a known job class.
7. Incumbent or fallback owner is ready to resume control.
8. Kill switch, credential revocation and route closure are tested.
9. Rollback restores a known-good state and preserves evidence.

10. Chronicle records decision, dissent, conditions and requalification trigger.

D.7. Release evidence index

A publication-grade succession package should include the following directory structure:

```
mission/  
  constitution/  
  baseline/  
  alternatives/  
  rights-and-risk/  
gym/  
  environment-spec/  
  source-or-binaries/  
  scenarios-public/  
  conformance/  
foundry/  
  candidate-manifests/  
  formation-results/  
  adversarial-results/  
proof/  
  preregistration/  
  protected-case-manifest-commitment/  
  run-receipts/  
  intervention-ledger/  
  cost-ledger/  
  scorecard/  
admission/  
  decision-record/  
  authority-envelope/  
  credentials-policy/  
operations/  
  canary-plan/  
  telemetry/  
  incidents/  
  rollback-and-revocation/  
chronicle/  
  entries/  
  lineage/  
  requalification/  
release/  
  manifest.json  
  SHA256SUMS.txt  
  README.md
```

Protected case content should not be placed in a broadly distributed release; the package can include commitments, custody attestations and access-controlled references.

D.8. Quality-assurance gates

Table 18: Release QA gates.

Gate	Pass condition
Schema QA	All required fields validate; identifiers and references resolve.
Environment QA	Reset, step, action rejection, endings, budgets and receipts pass conformance.
Security QA	No reusable secrets in candidate context; egress, credentials, dependencies and logs reviewed.
Evidence QA	Receipts are complete, hash-consistent, signed and attributable.
Statistical QA	Analysis reproduces from receipts; uncertainty and exclusions match pre-registration.
Authority QA	Every permitted action maps to an explicit admission clause and infrastructure rule.
Rollback QA	Known-good restore, fallback ownership and evidence preservation are demonstrated.
Language/UI QA	Proposed, verified, contested, invalidated and expired evidence are visually distinct.
Package QA	Clean-room extraction, manifest verification and SHA-256 comparison succeed.
Governance QA	Accountable decision, dissent, expiry and revocation authorities are recorded.

D.9. Decision memorandum template

Every admission decision should answer:

1. What exact mission and candidate release are being decided?
2. Which incumbent and credible alternatives were compared?
3. What is the protected lower-bound Mission Alpha?
4. Which Betas, tail risks and basis risks remain?
5. What is the resulting Sovereignty Spread and sensitivity?
6. Which hard gates passed or failed?
7. What dissent or unresolved proof debt remains?
8. Why is the chosen authority level the least sufficient level?
9. What expires, triggers requalification or causes immediate revocation?
10. Who is accountable for the decision and fallback?

D.10. Operating dashboard

The production dashboard should show, at minimum:

- exact operating release and Authority Envelope;

- ▶ evidence freshness and next expiry;
- ▶ current job class, budget, rate and exposure;
- ▶ mission outcomes and comparison baseline;
- ▶ human rescue and verifier disagreement;
- ▶ drift, basis-risk residual and incident state;
- ▶ rollback readiness and fallback owner;
- ▶ latest requalification result and Chronicle lineage.

It should not collapse this information into one reassuring percentage.

E. Glossary

Term	Canonical meaning
Accountable admission	External institutional act that converts proof into a bounded operating decision and exact Authority Envelope.
Action gateway	Infrastructure service that validates and executes typed actions without exposing reusable credentials.
Authority Envelope	Machine-enforceable specification of the exact actions, objects, budgets, approvals, jurisdictions, monitoring, duration, rollback and revocation conditions granted to one release.
Bounded AGI Job	Finite mission work unit with exact goal, inputs, outputs, actions, budgets, permissions, verifier, receipt and fallback.
Best credible alternative	Strongest operationally available or reasonably procurable human, general, specialist, vendor or hybrid alternative against which the candidate must be compared.
Canary	Limited real-world operation designed to test transfer while containing and reversing effects.
Chronicle	Governed institutional memory of valid capabilities, failures, decisions, conditions, lineage, incidents and invalidations.
Complete burden	All model, data, integration, operation, human rescue, switching, monitoring, rights, risk, rollback and authority costs.
Critical constraint	Non-compensable prohibition or ceiling whose violation makes a candidate ineligible or triggers immediate stop.
Environment capture	Condition in which the candidate, provider or formation team controls the authoritative representation of reality against which it is judged.
Fresh Proof	Protected, independent, comparative evaluation of a frozen release on unseen representative work.
Formation Gym	Environment visible to the formation team for development, replay, architecture search and repair.
Fresh-Proof Gym	Protected environment used for promotion evidence, with hidden cases, frozen metrics and independent custody.
Gym basis risk	Difference between the value measured in the Gym and the value realized in the real mission.

Term	Canonical meaning
Gymmability	Degree to which a mission can be reset, observed, bounded, measured, adversarialized, contained and tested for transfer.
Human rescue	Human intervention needed to correct, complete, interpret or operate the candidate beyond the designed workflow.
Incumbent	Current policy, process, team or operating architecture that the candidate may replace.
Mission Advantage Gradient	Direction over architecture or experiment choices that most increases expected Sovereignty Spread per unit of proof capital.
Mission Alpha	Comparative excess mission value over the best credible alternative.
Mission Beta	Sensitivity of mission advantage to systematic factors, regimes, providers, data, humans, integration or verifier dependencies.
Mission Constitution	Versioned definition of mission, beneficiary, alternatives, information, work, value, constraints, proof capital, evidence, authority ceiling and expiry.
Mission Gym	Controlled, executable and versioned representation of mission states, observations, actions, transitions, outcomes, constraints, scenarios and evaluation.
Mission-Sovereign Successor	Proven Specialist ASI after accountable admission, identity, Chronicle linkage, monitoring, rollback, expiry, revocation and scoped authority.
Proof capital	Bounded budget of cases, data, compute, expert time, experiments and risk allocated to reducing a succession decision's uncertainty.
Proof debt	Material uncertainty, missing evidence, evaluator weakness, contamination or unresolved assumption that remains after apparent performance.
Proof receipt	Versioned evidence artifact emitted by an episode, including identity, scenario, observations, actions, interventions, value, constraints, costs, endings and verifier results.
Requalification	Fresh evidence process triggered by time, drift, incident or material change to model, data, tools, workflow, verifier, provider, mission or authority.
SEIZE Underwriting	Named GoalOS layer that converts a Successor Manifold opportunity into a bounded thesis, anti-thesis, impairment case, proof plan, risk reserve and stop conditions.
Specialist ASI	Complete candidate intelligence institution that proves mission-specific superiority over the incumbent and best credible alternatives on fresh work.
Sovereignty Spread	Risk-adjusted lower-bound Mission Alpha after complete burden, human rescue, Beta, tail risk, Gym basis risk, proof debt and authority burden, plus justified option value.
Successor Foundry	Isolated formation system that generates and improves competing complete candidate architectures.

Term	Canonical meaning
Successor Manifold	Structured design space of credible architectures and transaction modes for retaining, repairing, renting, forming, partnering, acquiring, hedging, reserving or retiring mission capacity.
Successor Ω	Mission-Sovereign Successor bound to a governed, non-self-authorizing process for Chronicle, requalification and production of its next successor.
Typed ending	Explicit classification of success, mission failure, critical failure, termination, truncation, operator stop or authority expiry.
Verifier	Independent method or institution that determines whether mission claims, evidence and constraints are supported.
Viability kernel	Set of mission states from which at least one admissible policy can keep the system within constitutional constraints for the required horizon.

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